



Particle Detection

Timepix pixel detectors (part II)

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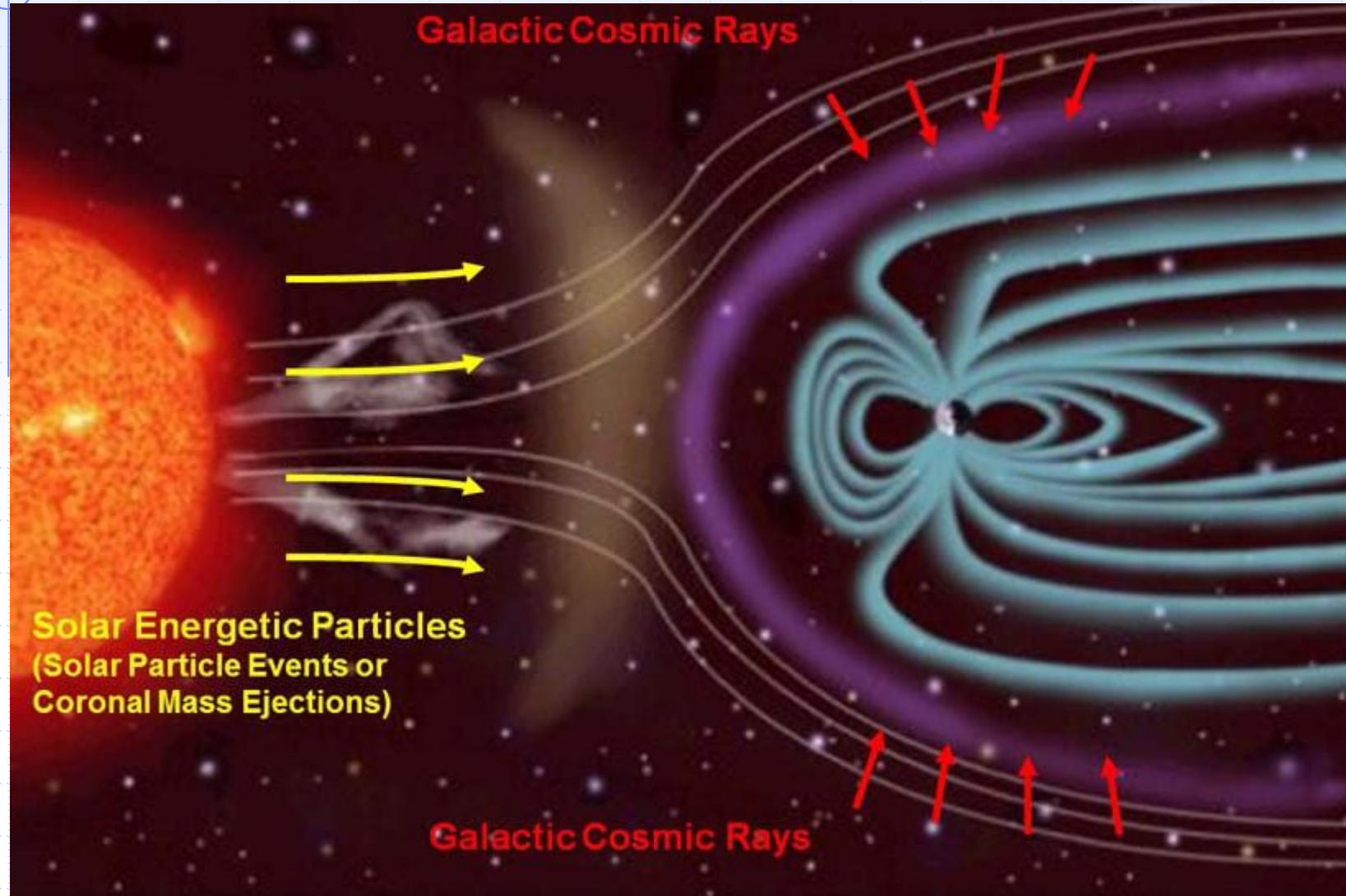
SAGI 2025, Quy Nhon
2025/07



Space and Particles



Space is a “microwave oven” filled with **mysterious** **damaging** and **deadly** energetic **particles**



Popular perception of space

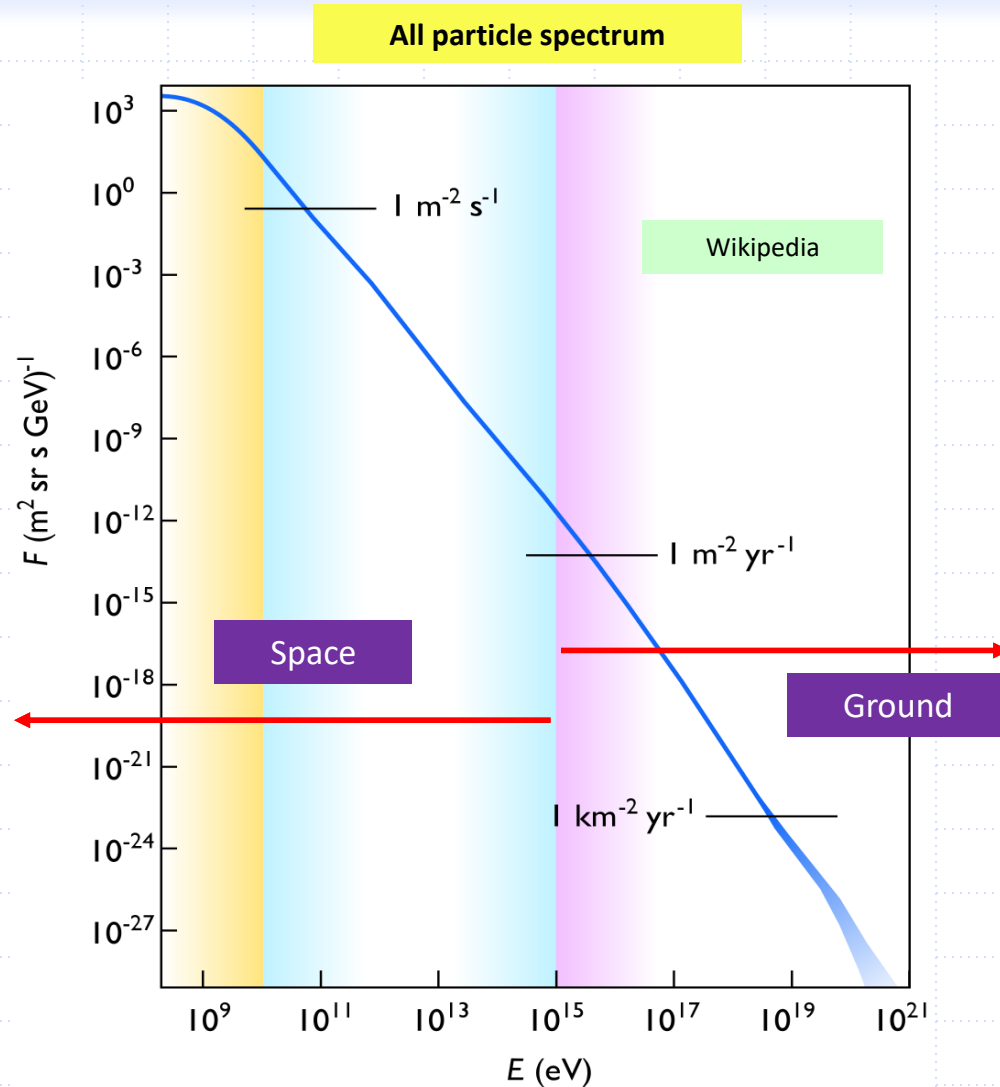
- No air
- Cold
- Weightless
- ~~Empty~~

The Earth is protected by its own magnetic shield (and its atmosphere)

1 day in space $\approx$ 1 year on Earth for radiation exposure, **even with spacecraft protection**



General Features of Cosmic Ray Observation



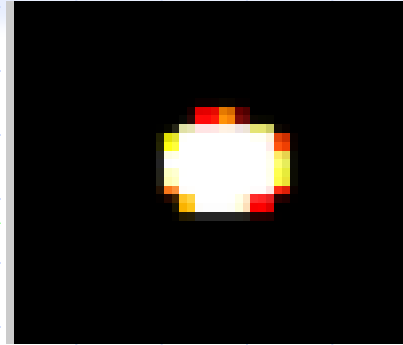
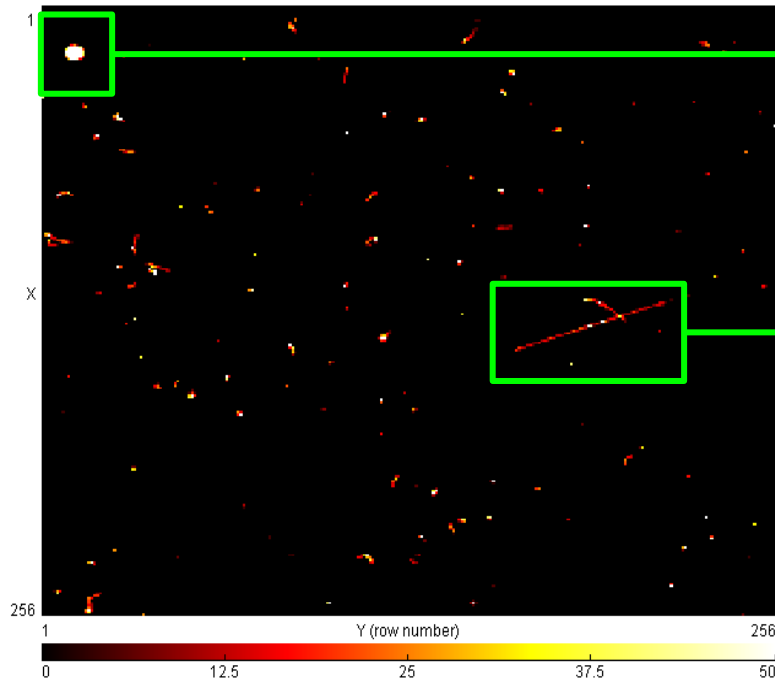
- **Roughly**, flux follows a power law $F(E) \propto E^{-\gamma}$, $\gamma \sim 2.7$ up to the “knee” at $\sim \text{PeV}$, then an “ankle” at $\sim \text{EeV}$
 - 30 orders of magnitude in flux
 - 12 orders in energy
 - Need **space** + ground experiments!
 - Need **different techniques of detection!**



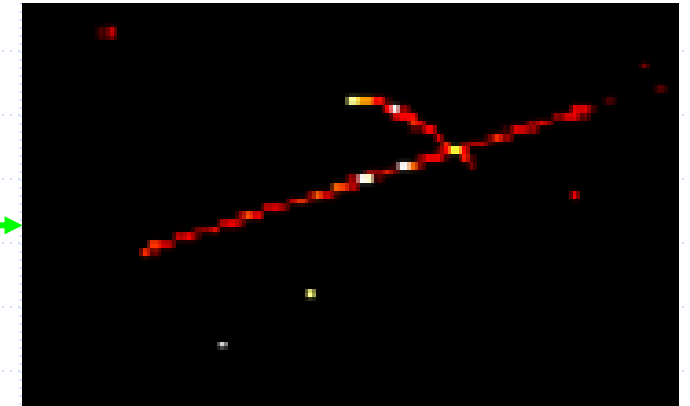
Typical observation of radiation background done by Timepix detector at the ground level



Exemplar of long term measurement
(exposition in tens of minutes)



- Detected alphas originates from decay of radon and its daughter products (count depends on concentration of radon in the air)



- Most of detected event are dots or curly tracks (corresponding to gammas)

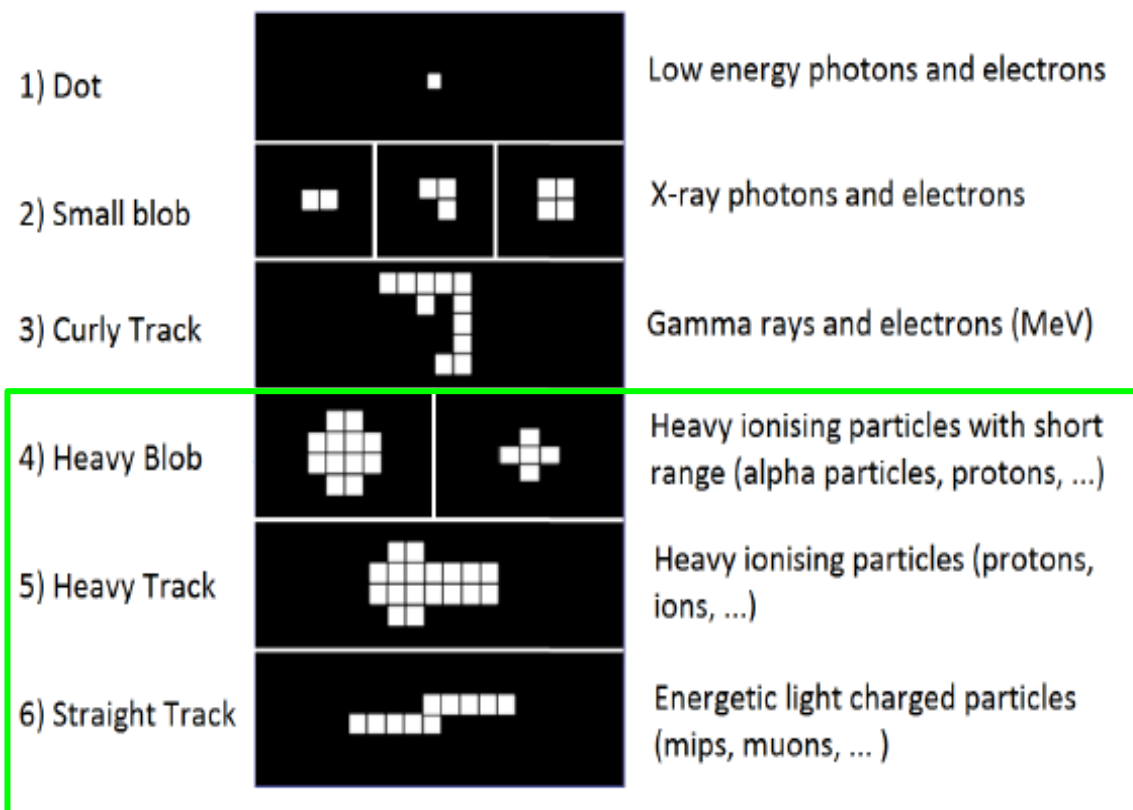
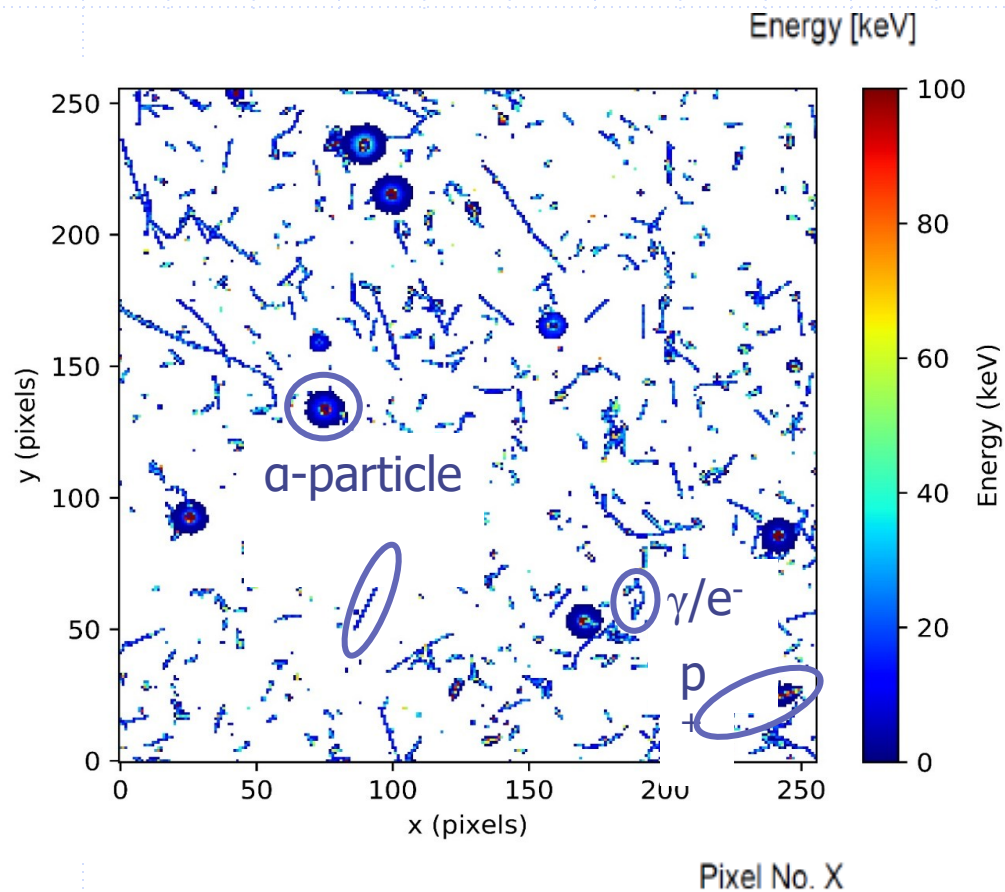
- Cosmic muons are also observable, however rarely

What we can see when getting to the higher altitude???



Remindfer of Timepix working principle: Pattern recognition

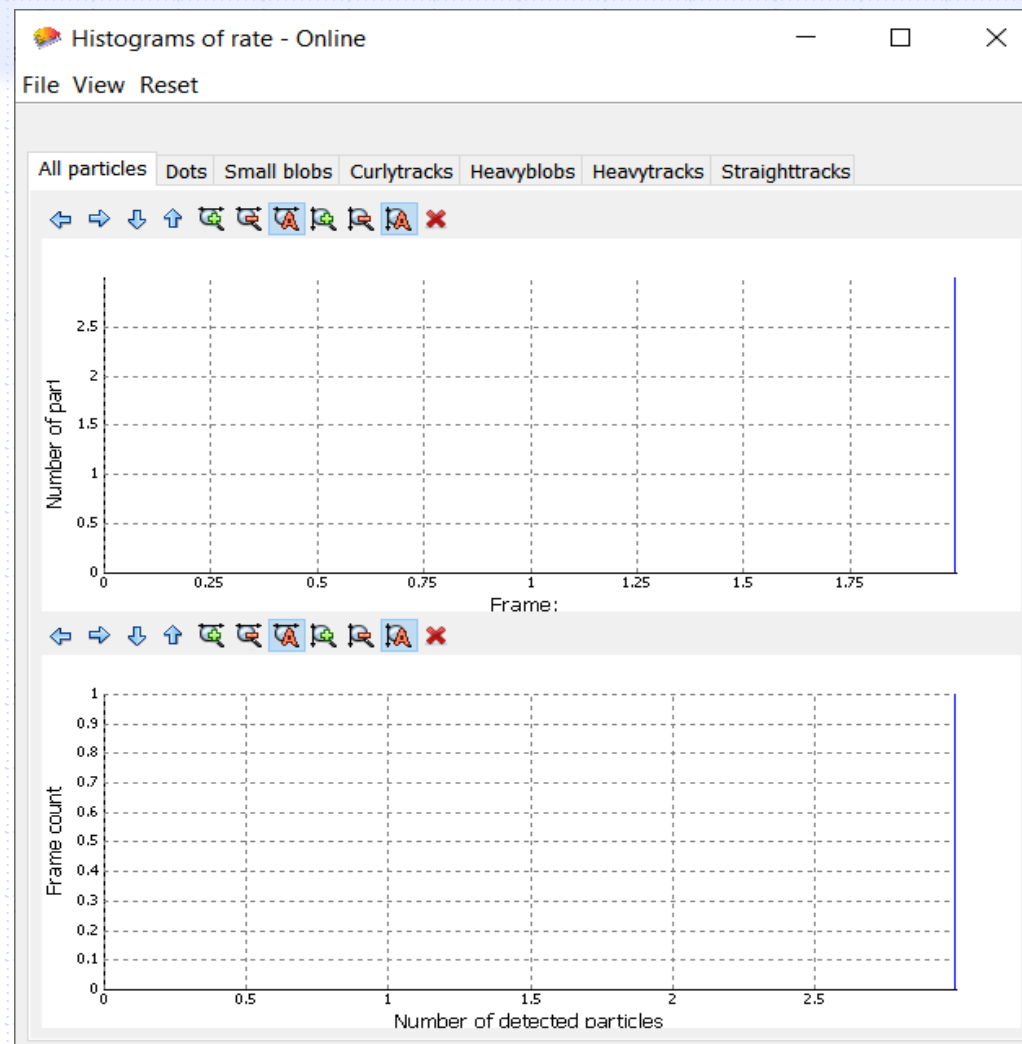
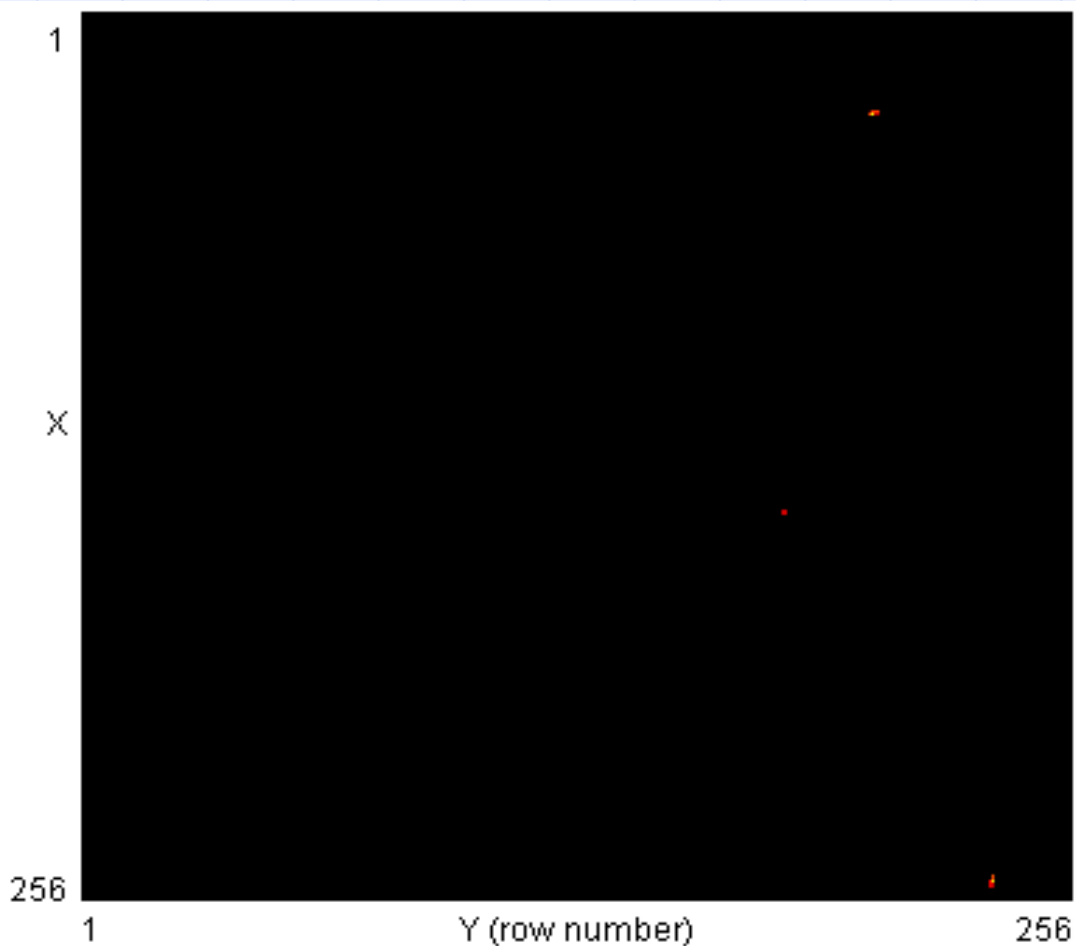
Pattern recognition together with dE/dX information allows determination of incident particle type and energy



T. Holy, et al. *Pattern recognition of tracks induced by individual quanta of ionizing radiation in Medipix2 silicon detector*, Nucl. Inst. Meth. Phys. Res. A, 591(1):287 – 290, 2008.



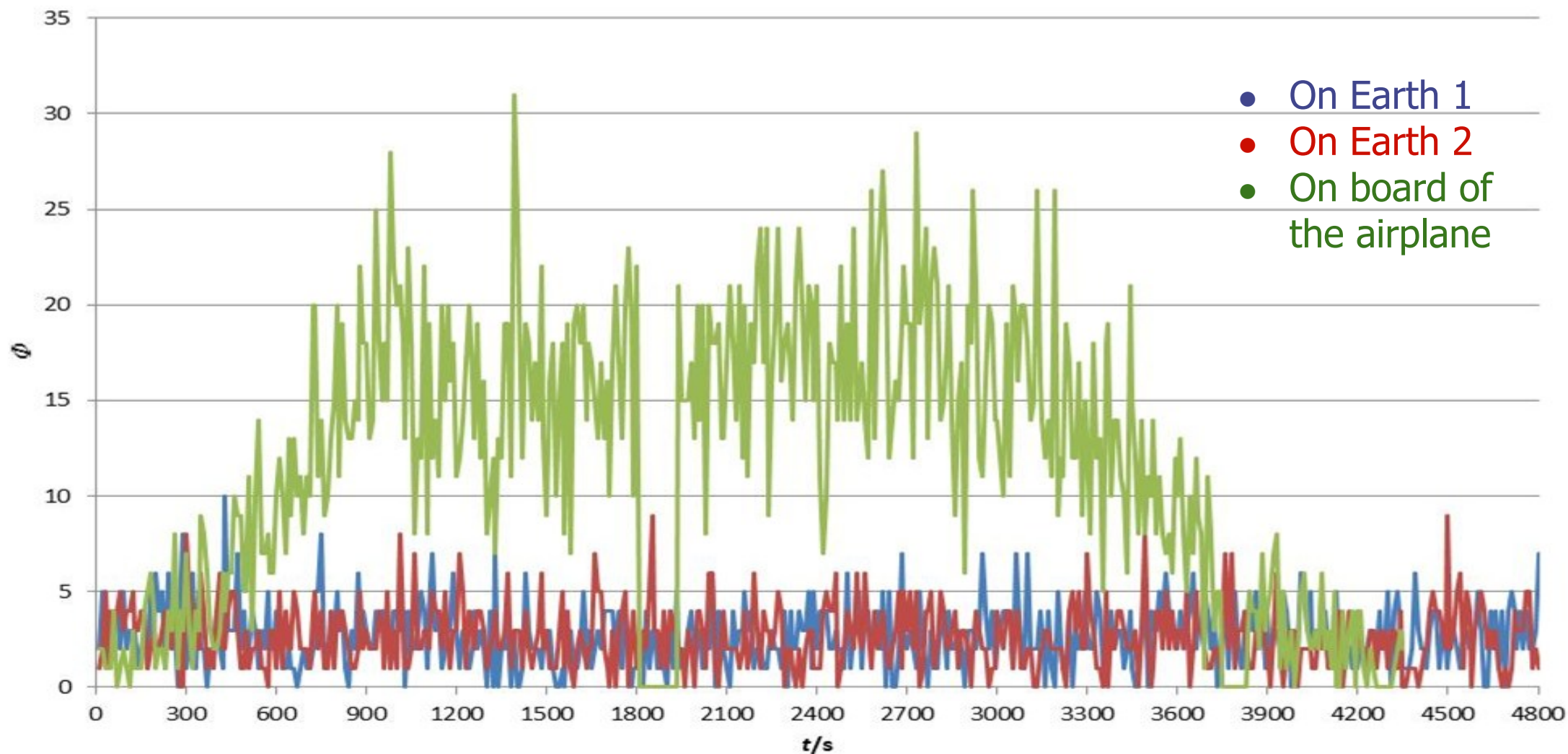
Radiation field observation on board of the airplane as seen by Timepix detector



- Recording at the flight from London to Prague
- Significant difference in field composition and intensity when flying or when landed...



Radiation intensity observed on **Earth** vs on board of the **airplane**



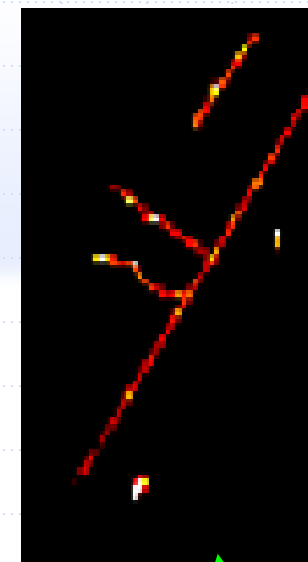
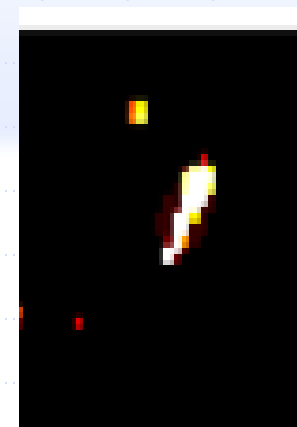
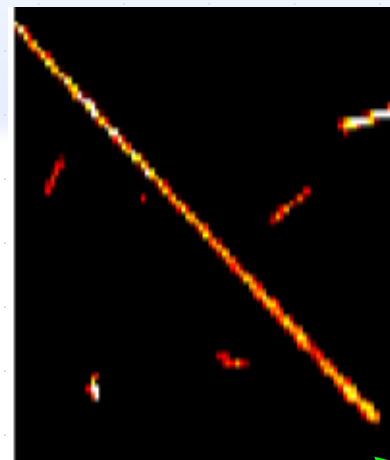
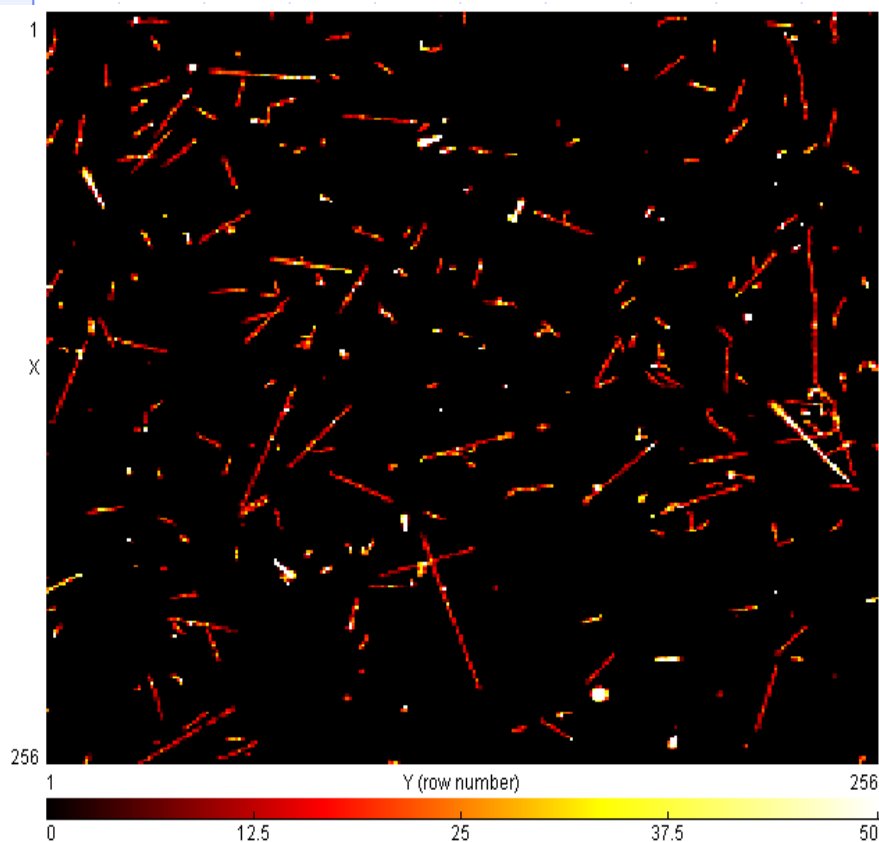
Particle flux Φ in dependence on time (with 10 s step)



Close look on airplane flight observation recorded by Timepix detector

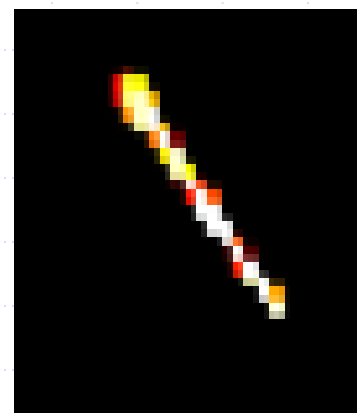


- Typical composition of radiation field at the standard flight altitude (commonly ~10000m)



- Notable high count of muons

- Occasional detection of heavy energetic ions



Close look on detected events

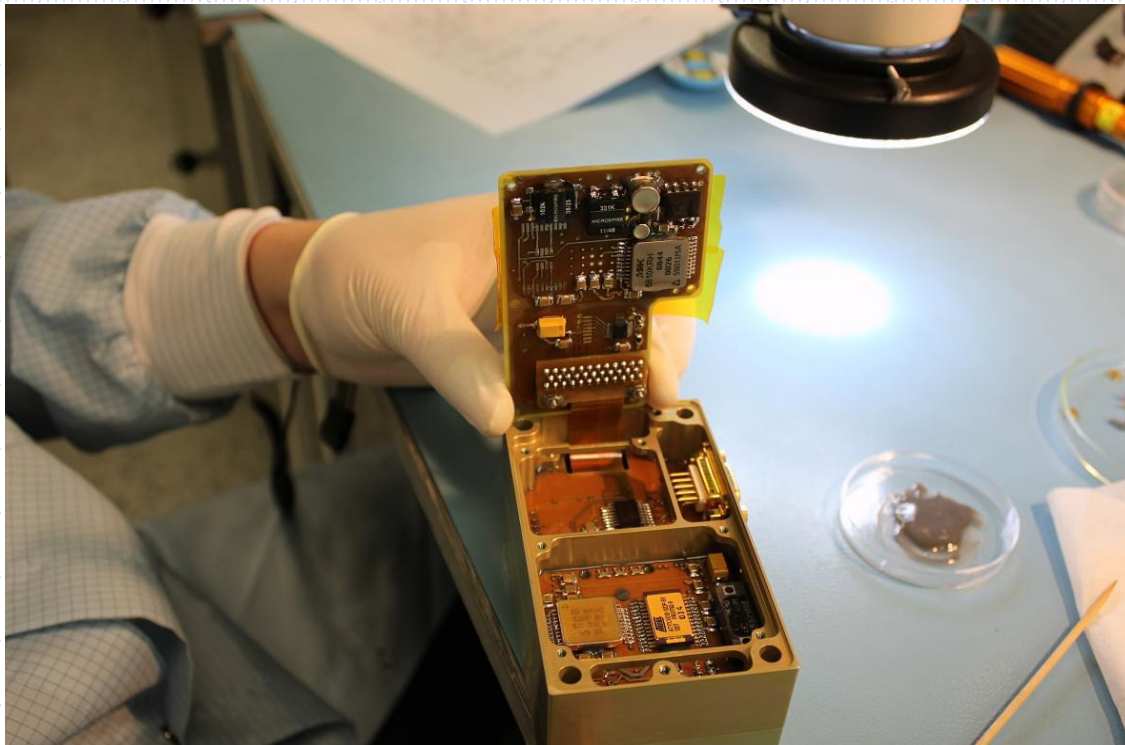


Dosimetry in space: **SATRAM** – ESA Proba-V satellite (*Space Application of Timepix Radiation Monitor*)



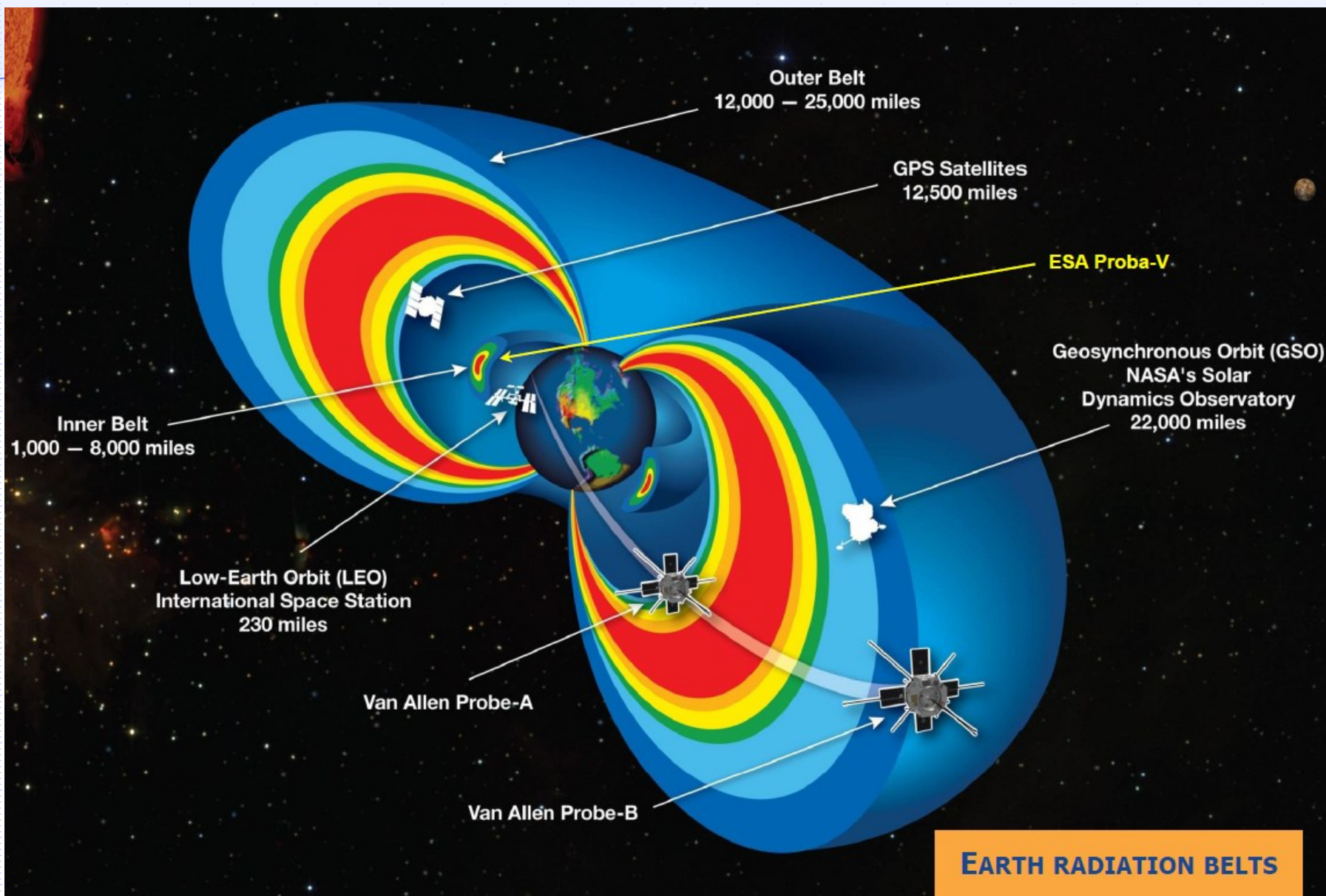
Characterization of mixed radiation field on low orbit of PROBA-V satellite

- ◆ Altitude ~ 800 km
- ◆ Timepix for the first time outside in the space
- ◆ Launched in May 2013





Proba-V/SATRAM on the earth orbit

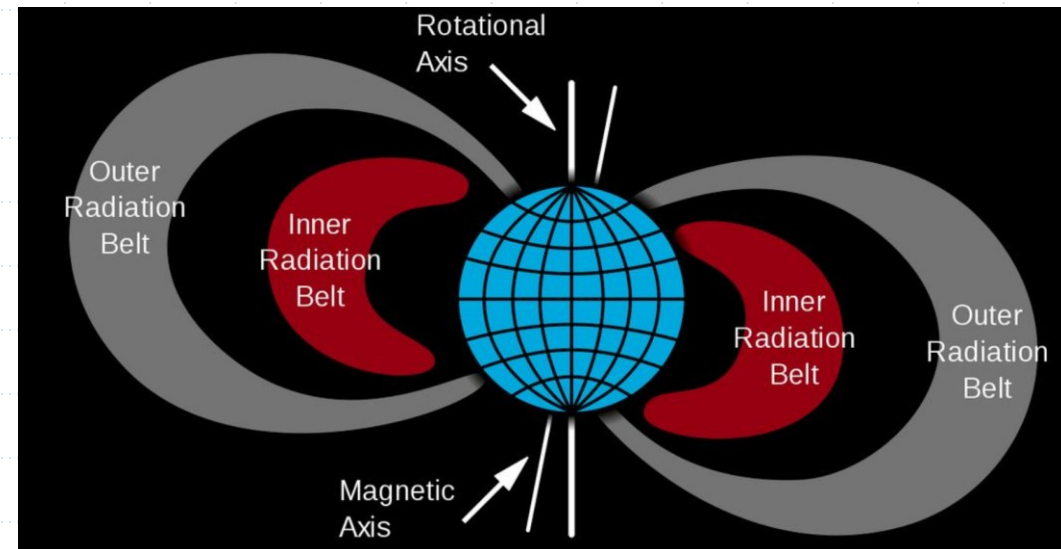
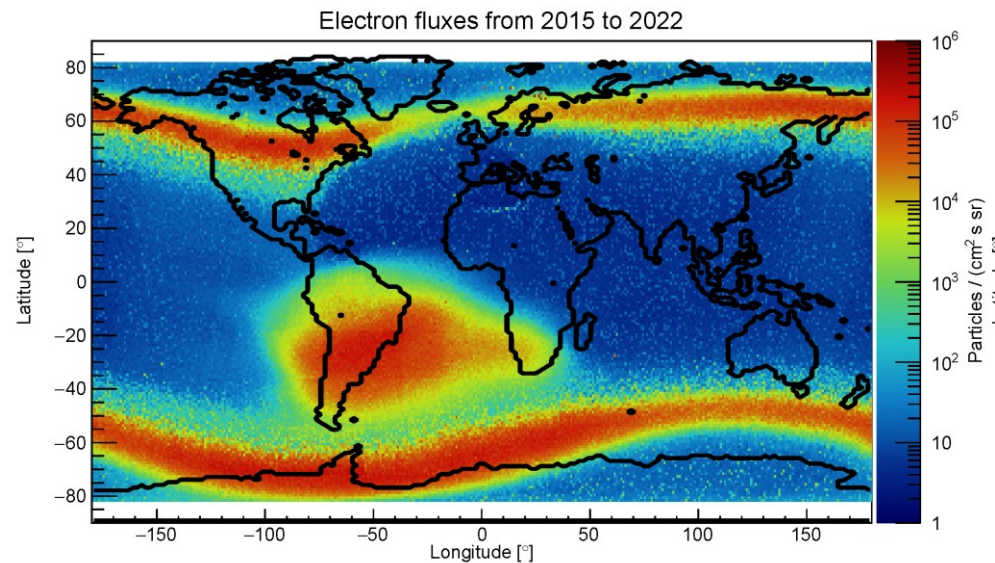


Proba-V satellite
altitude ~ 800 km



Radiation around Earth

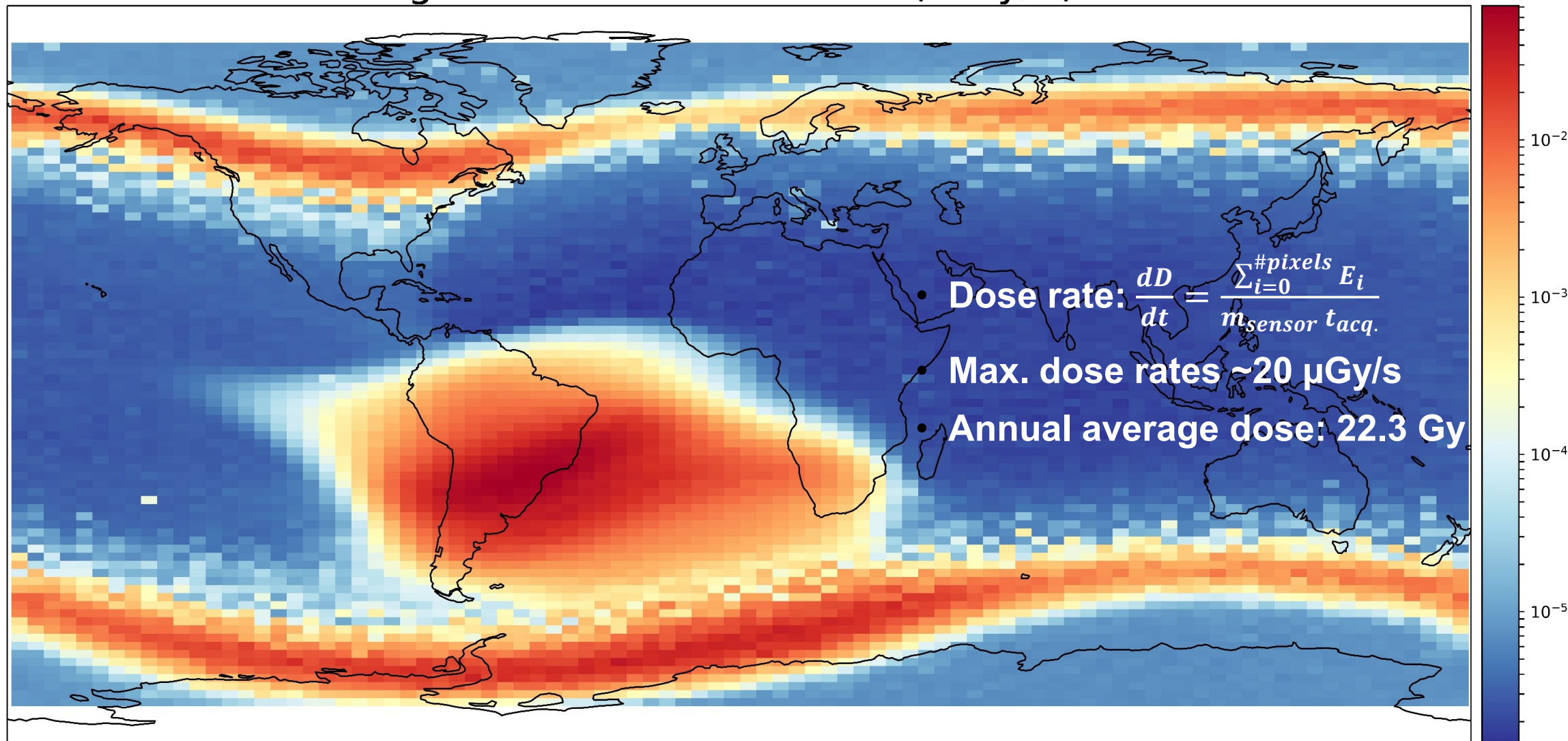
- ◆ In LEO, Van Allen belts create regions of high cosmic ray flux near poles and at South Atlantic Anomaly
- ◆ Polar regions are caused by magnetic field lines entering atmosphere *aurora borealis, northern lights*
- ◆ SAA caused by Earth rotational axis and magnetic axis not crossing in Earth's center, southern part of inner radiation belt enters the atmosphere



- Rotation axis $\neq$ magnetic axis
- Center of magnetosphere $\neq$ center of Earth

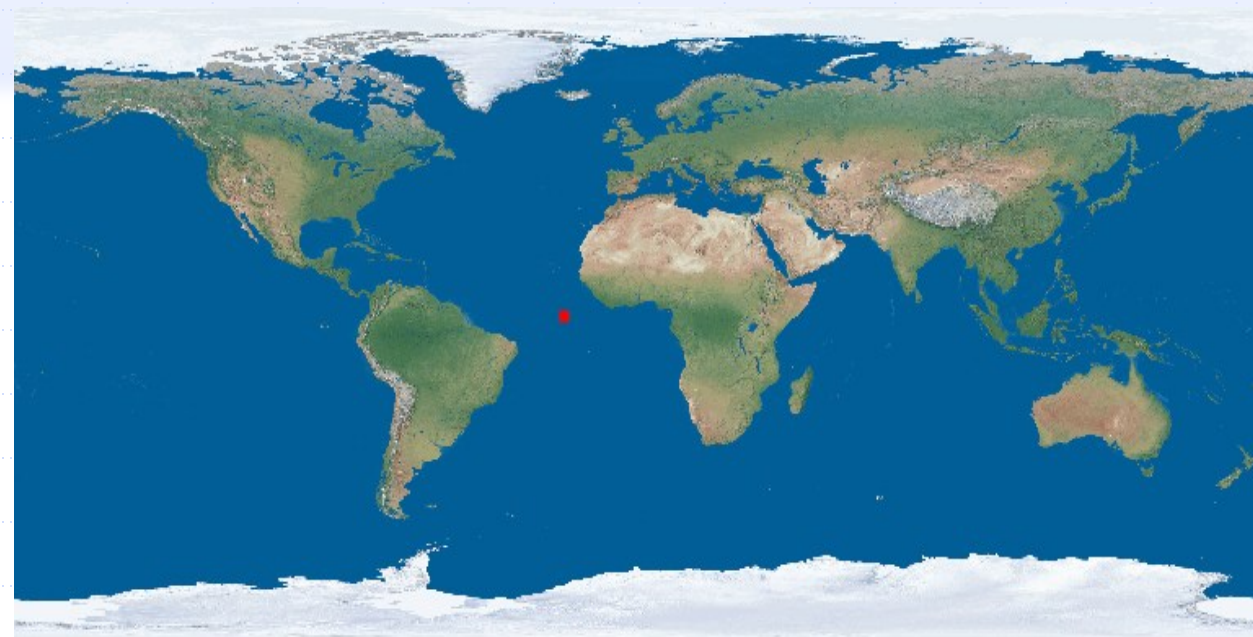
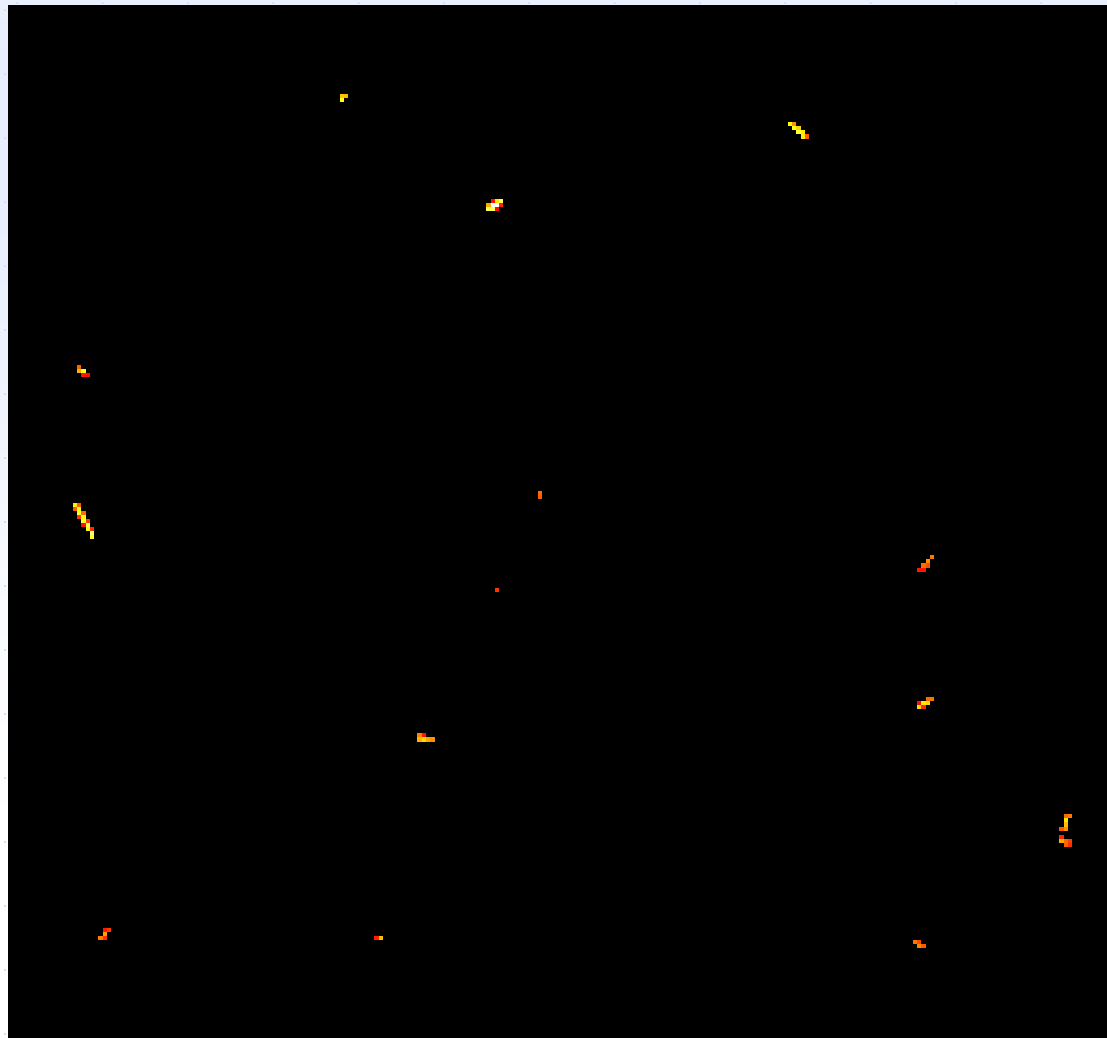


SATRAM - Average dose rate 2015-2018 (mGy/h) - Orbit: 820 km





SATRAM Radiation Observation on the Earth Orbit



- Radiation field as seen by Timepix detector

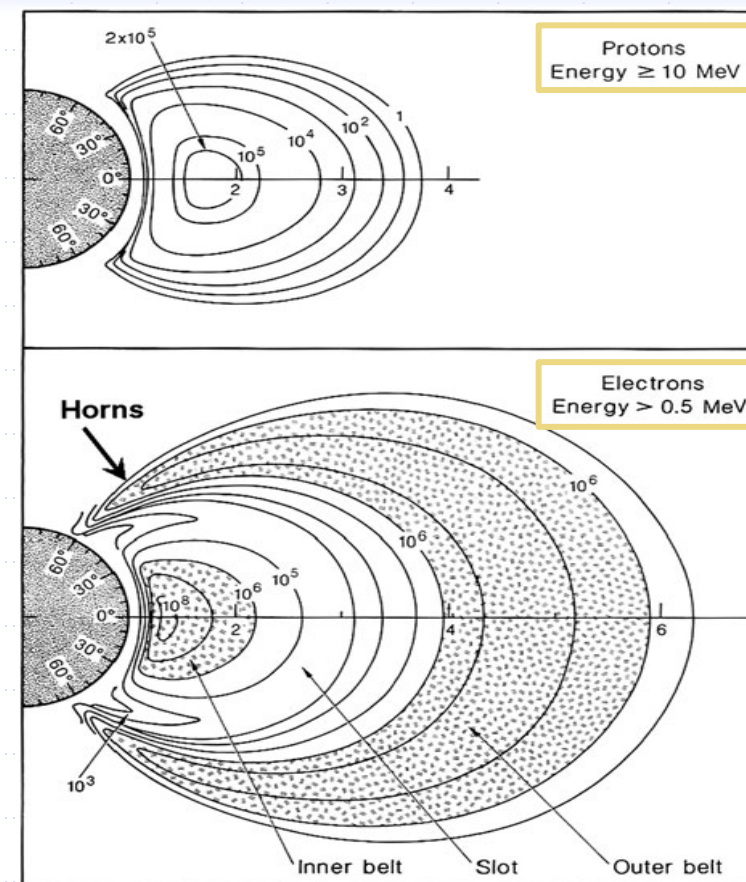
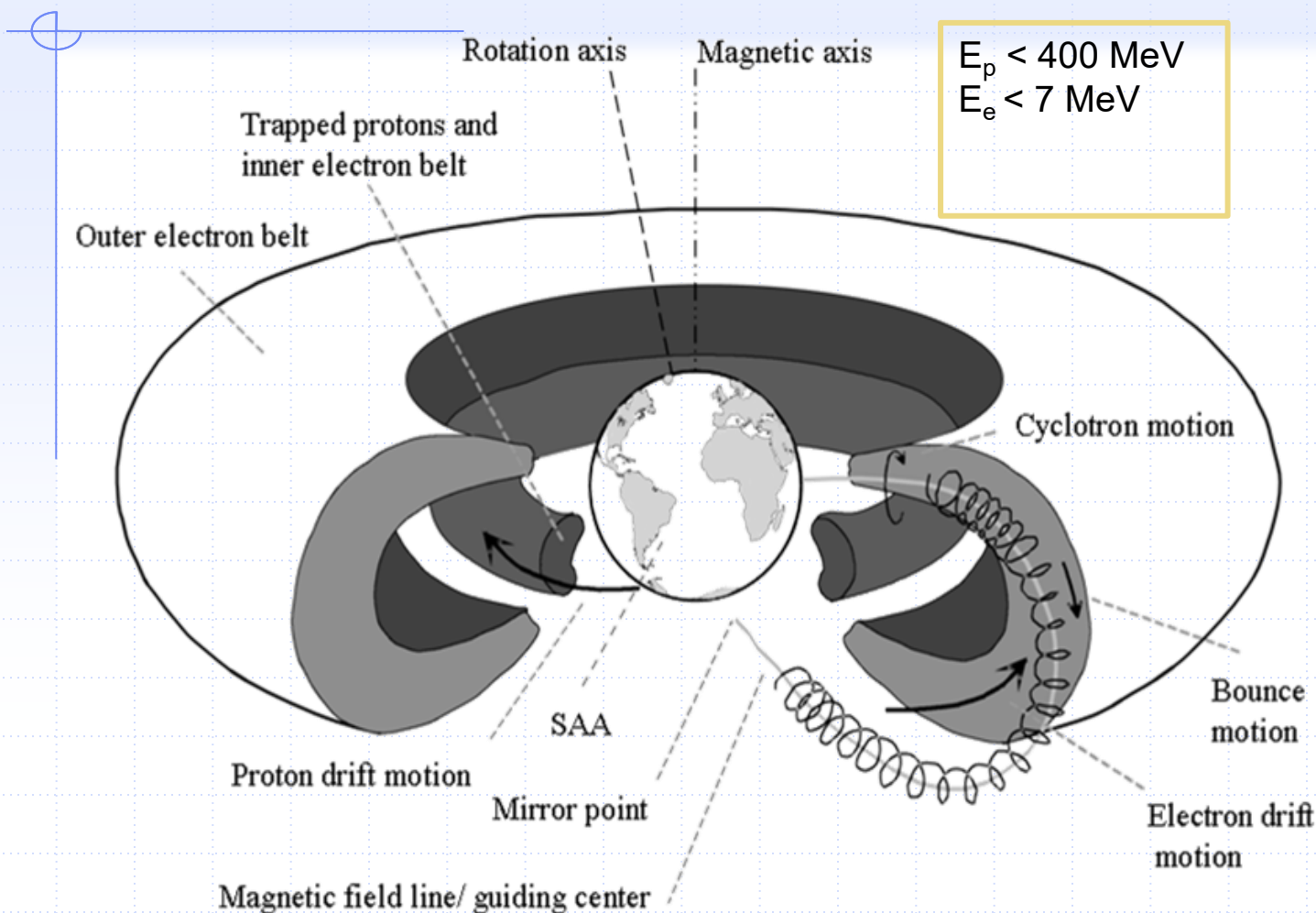
- Variation of radiation field composition in dependence on the earth orbit position
- Notable difference in field composition when compared to observations performed at ground level



Mission data available online at the web portal:
<https://satram.utef.cvut.cz/>



Radiation environment in LEO



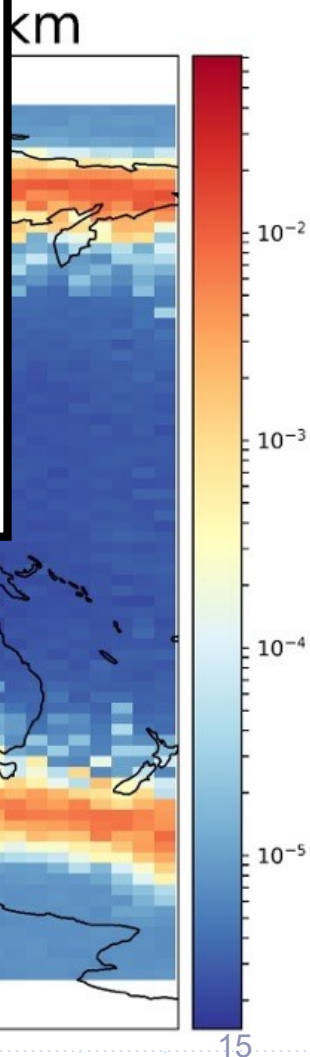
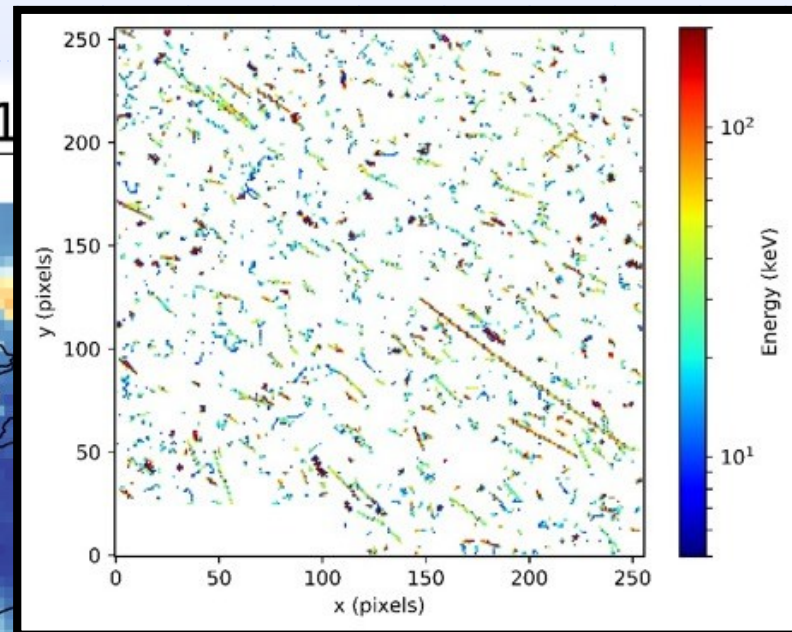
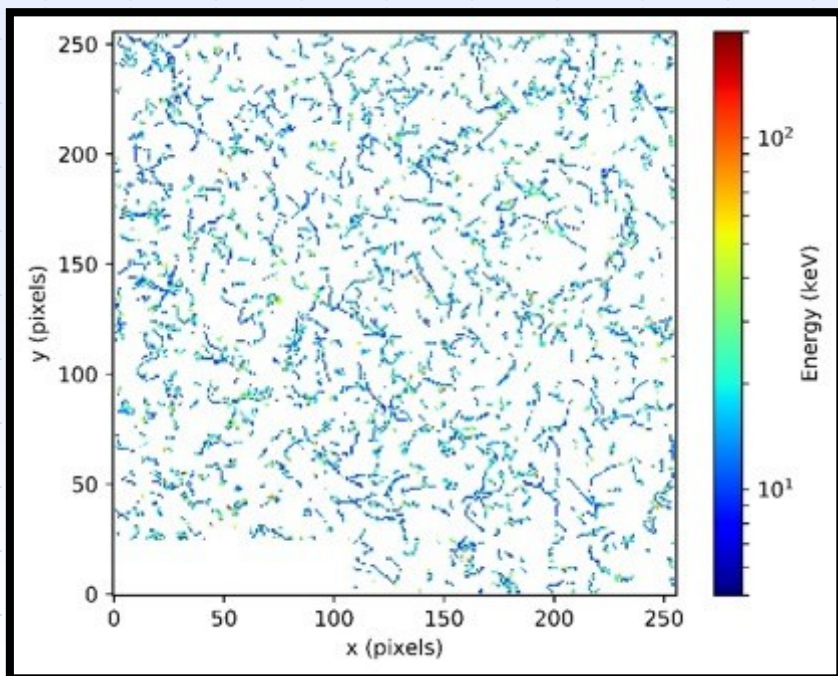
Mauk, B.H., Fox, N.J., Kaneal, S.G. et al. Space Sci Rev (2013) 179: 3.

Note> The term *LEO (Low Earth Orbit) region* is used for the area of space below an altitude of 2,000 km



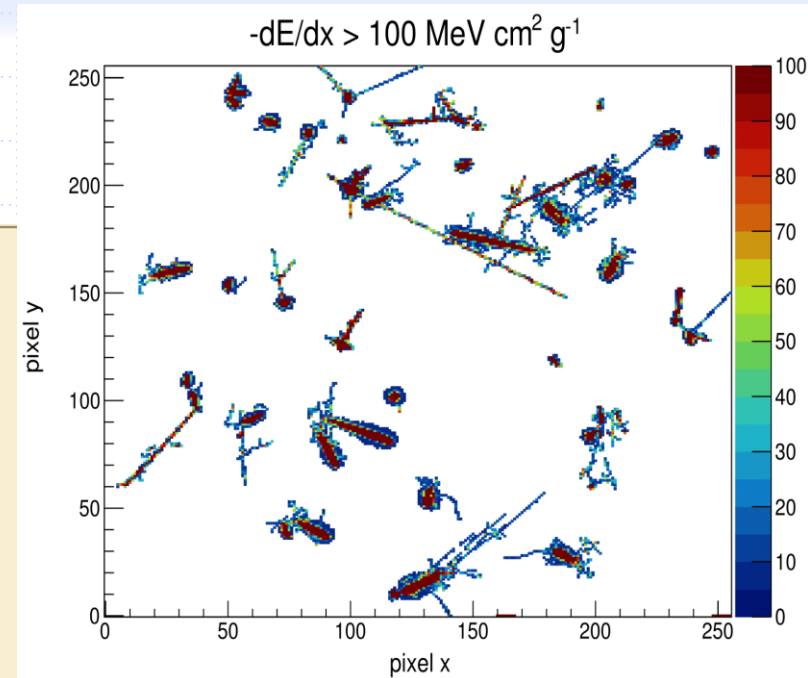
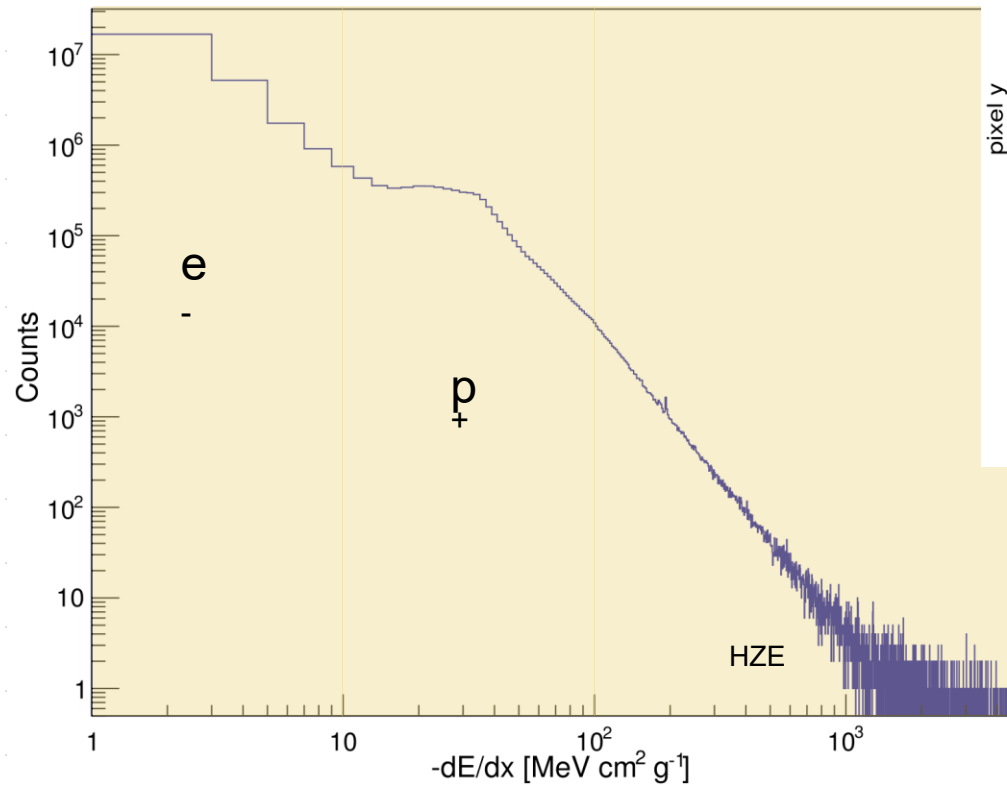
SATRAM

Focusing on composition of radiation field





dE/dX and particle classification

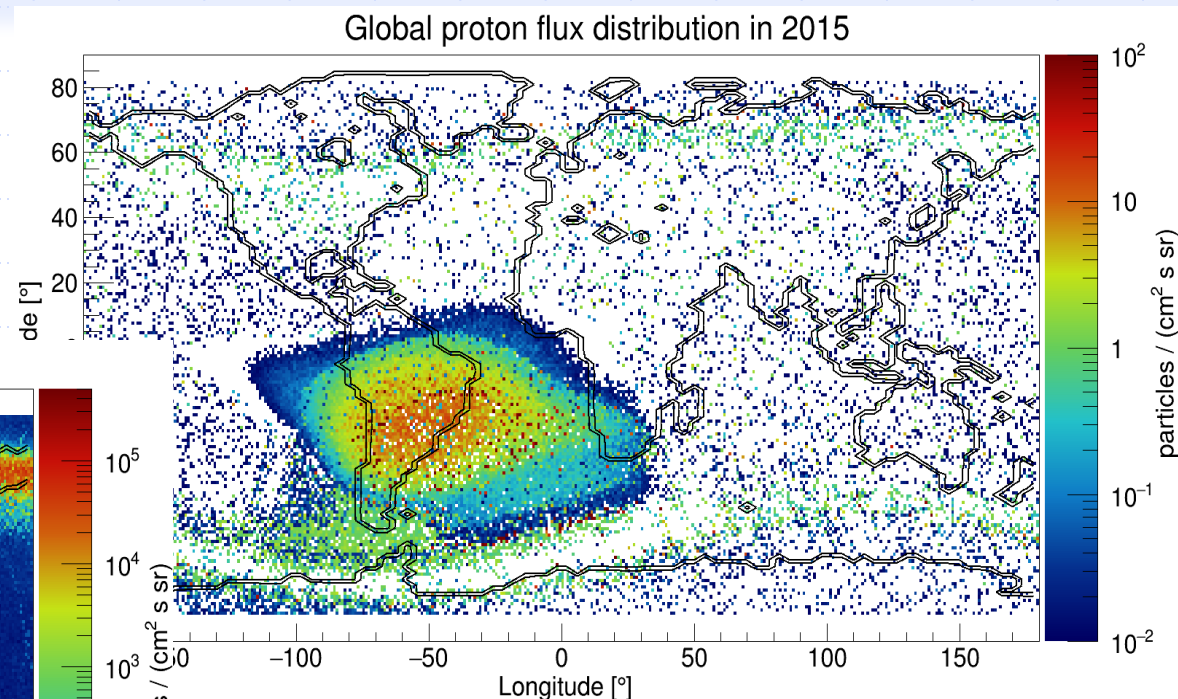
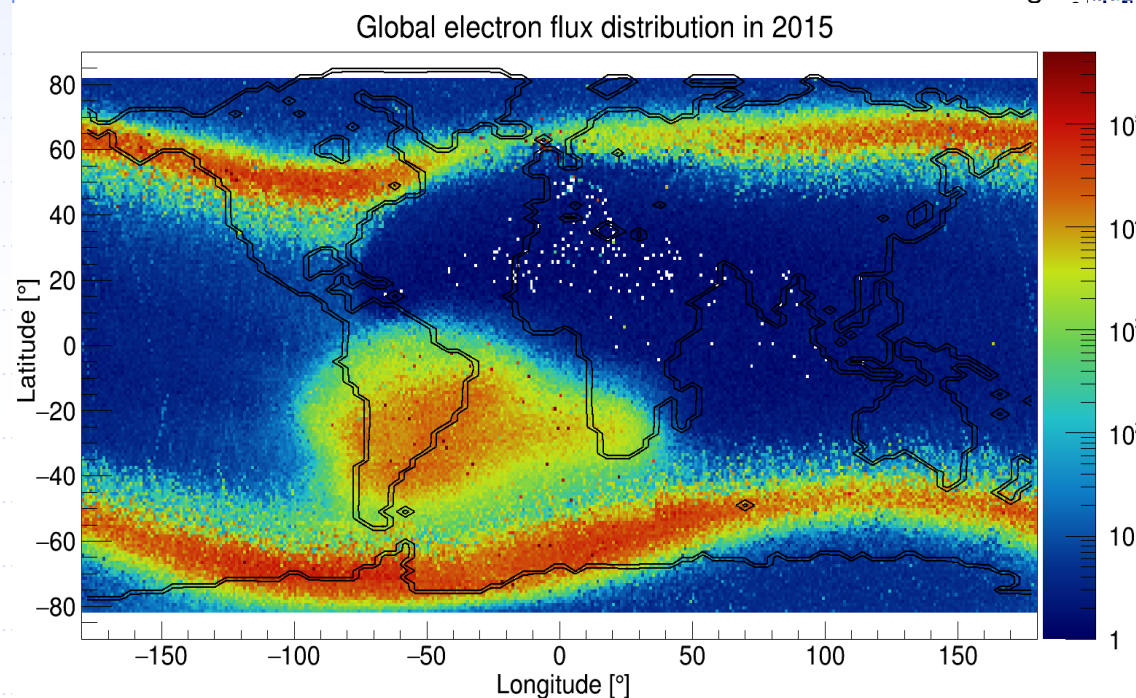


St. Gohl et al., "Study of the radiation fields in LEO with the Space Application of Timepix Radiation Monitor (SATRAM)", *Advances in Space Research* **63**, Issue 5, pp. 1646-1660, (2019).



Electron and proton flux maps

e^- fluxes 3 orders of magnitude larger than p^+ fluxes
→ Even small e^- misclassification distorts p^+ flux measurement



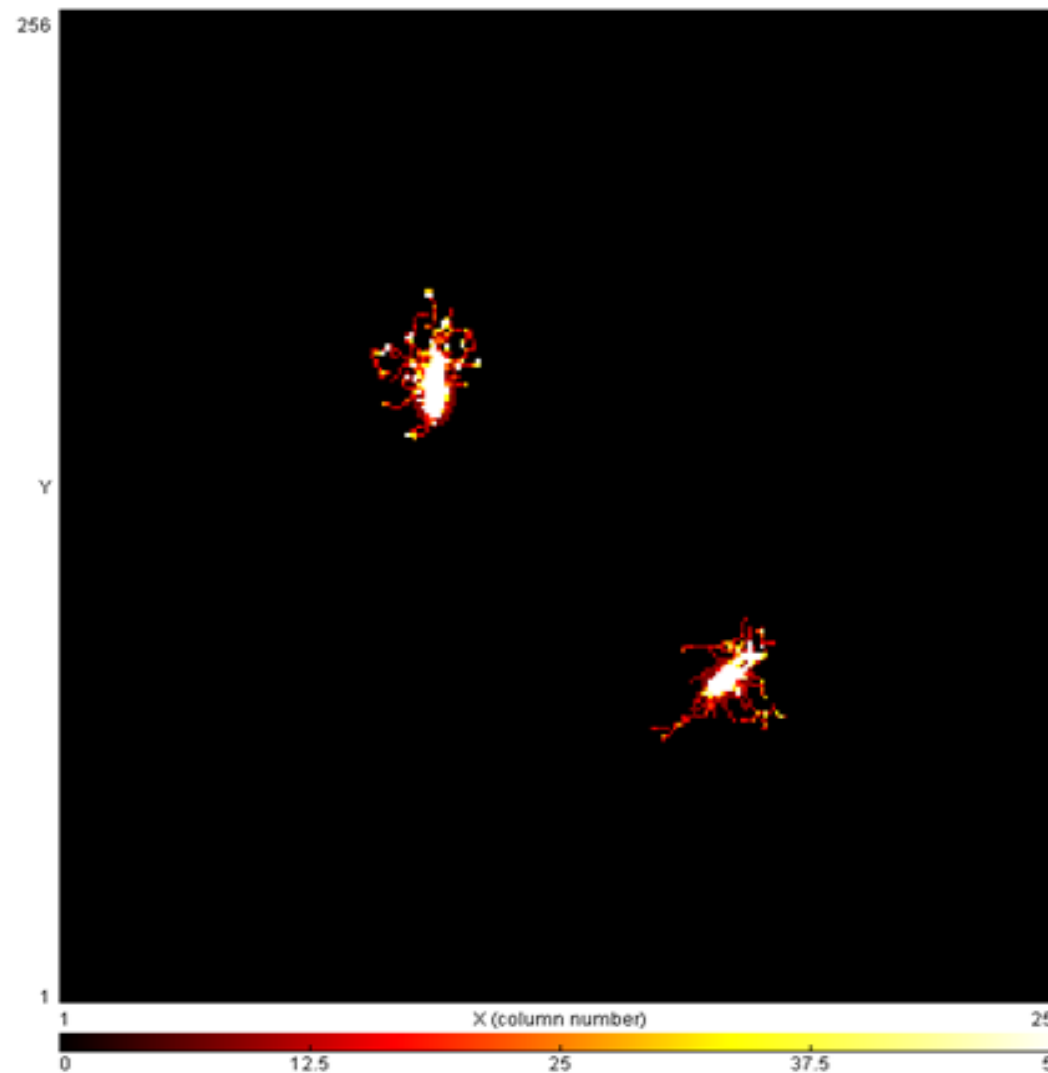
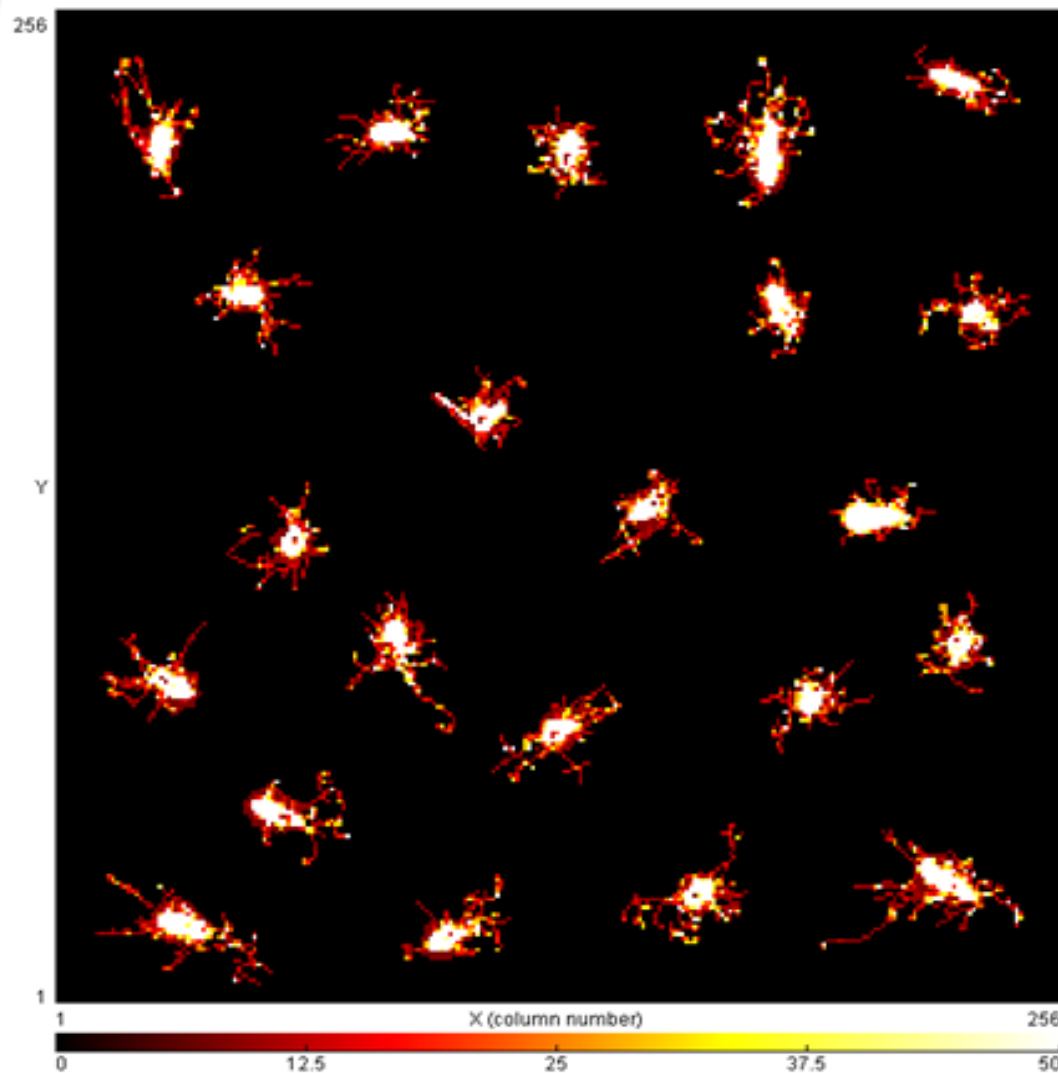
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Timepix/ESA Proba-V: LEO space radiation @ 820 km



HETPs: Highly energetic heavy charged particles (ions) → HZE's

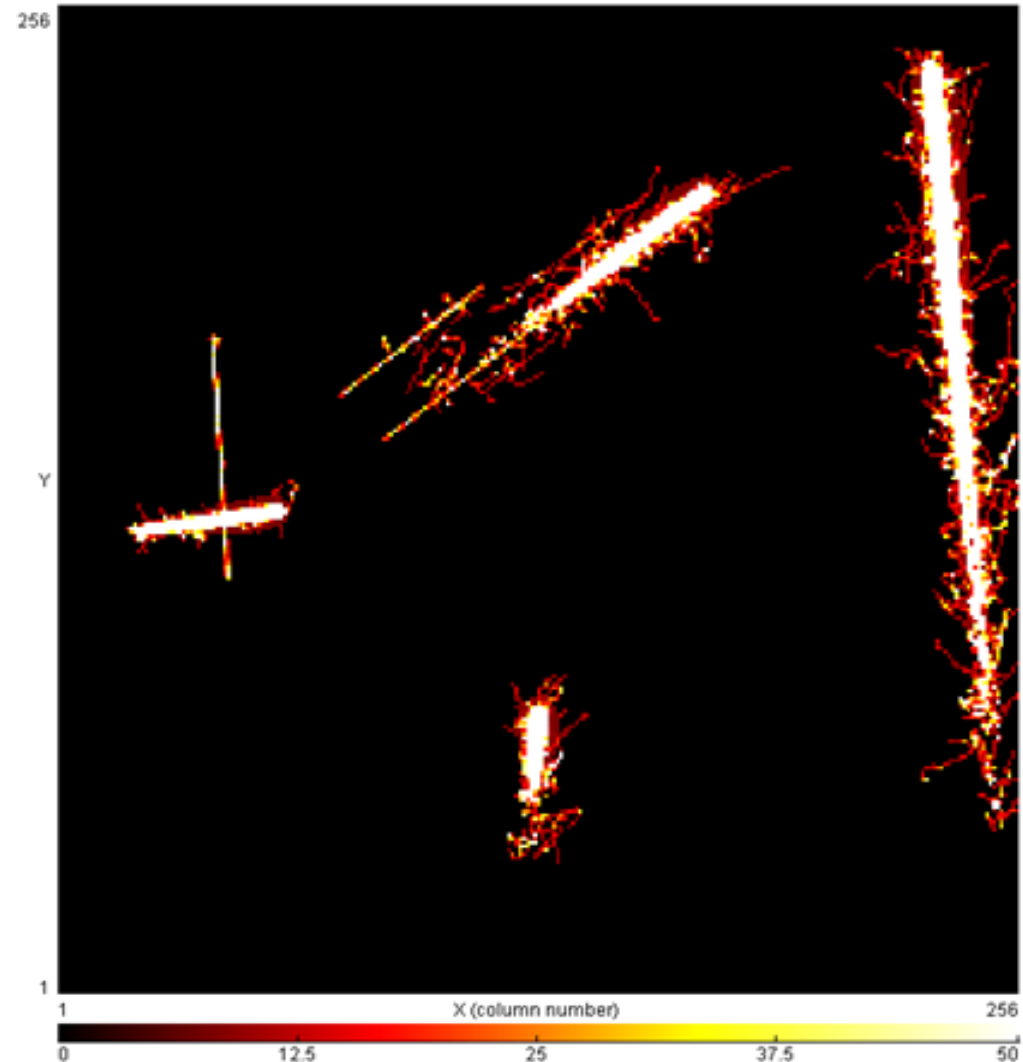
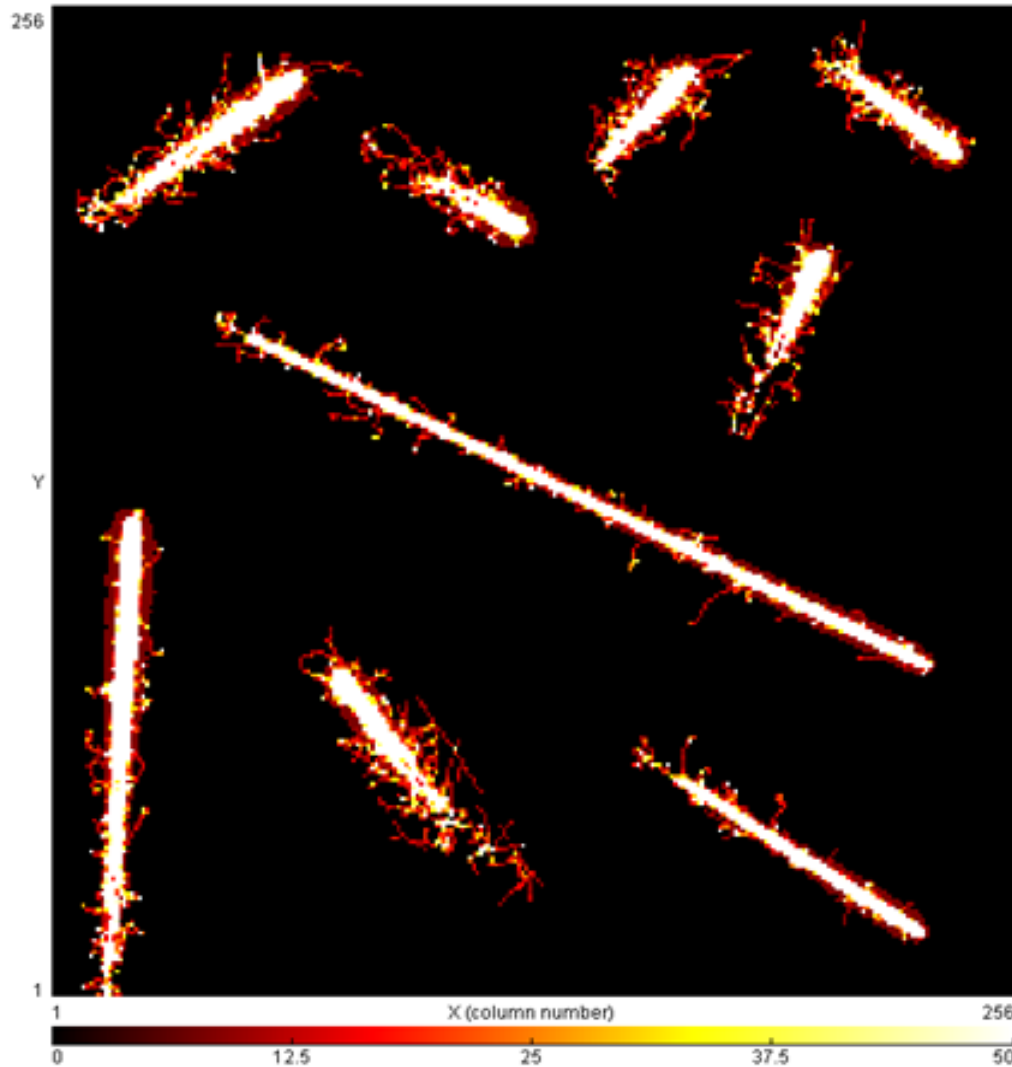




Timepix/ESA Proba-V: LEO space radiation @ 820 km



HETPs: Highly energetic heavy charged particles (ions) → HZE's

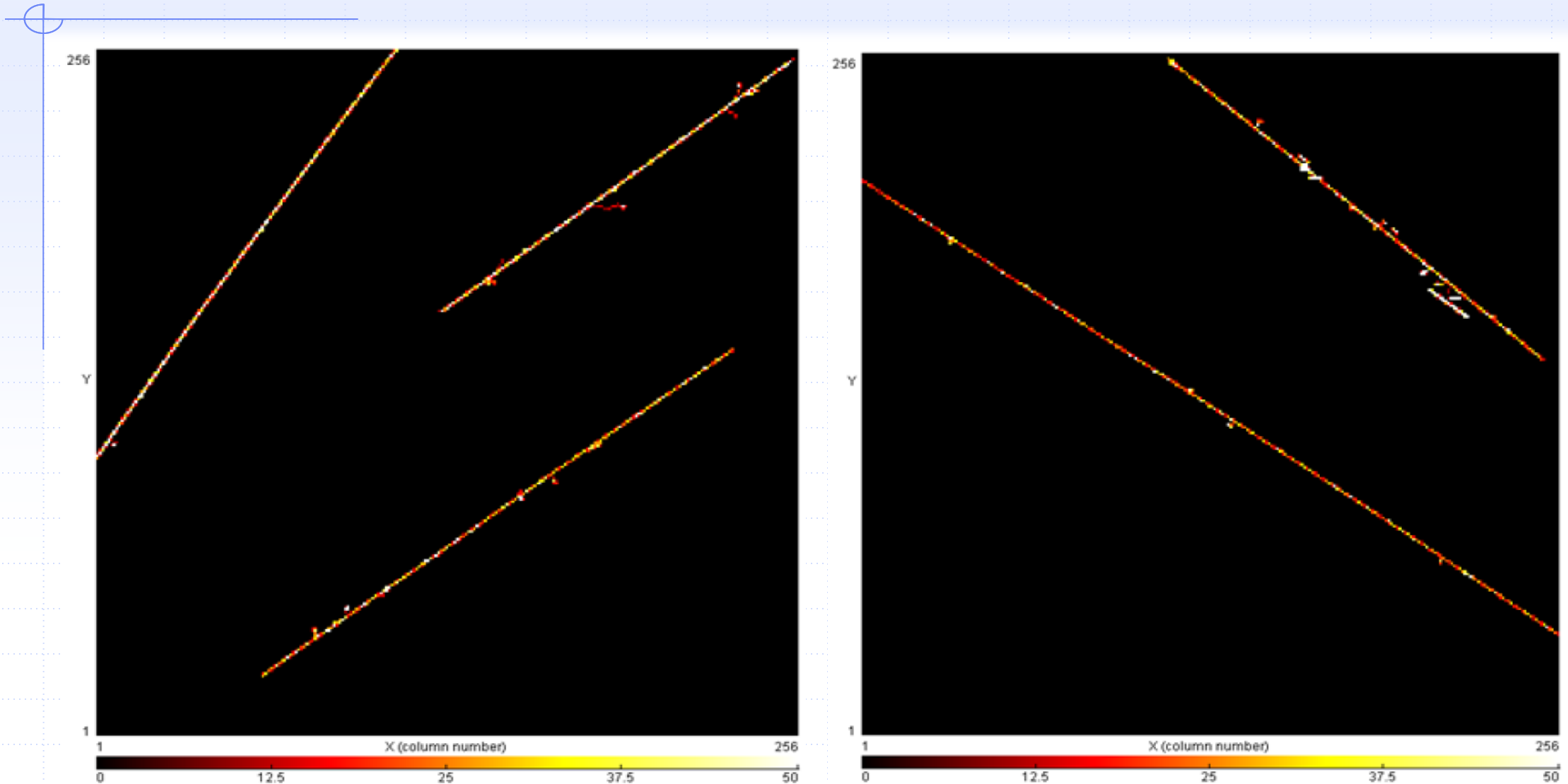




Timepix/ESA Proba-V: LEO space radiation @ 820 km



LETPs: Energetic light charged particles (I)

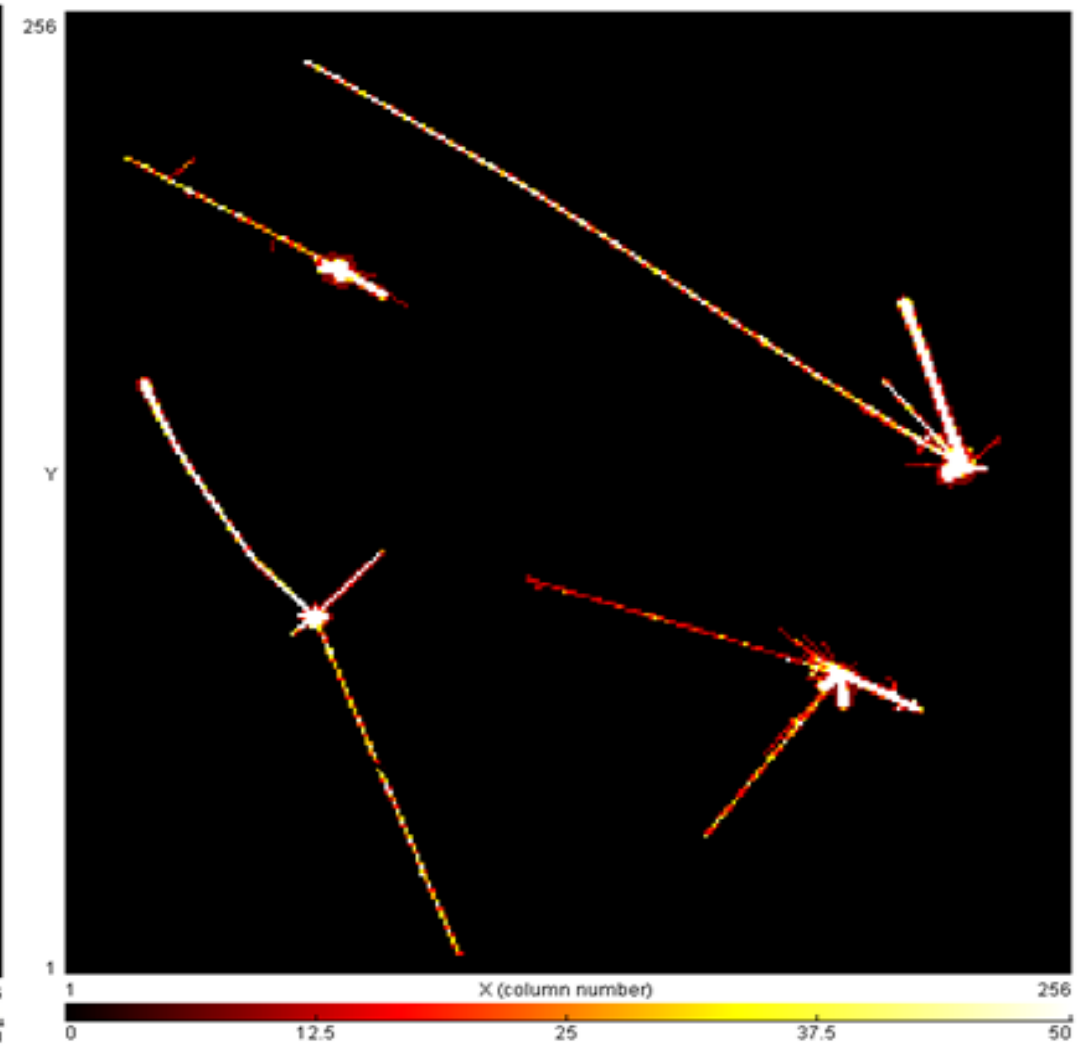
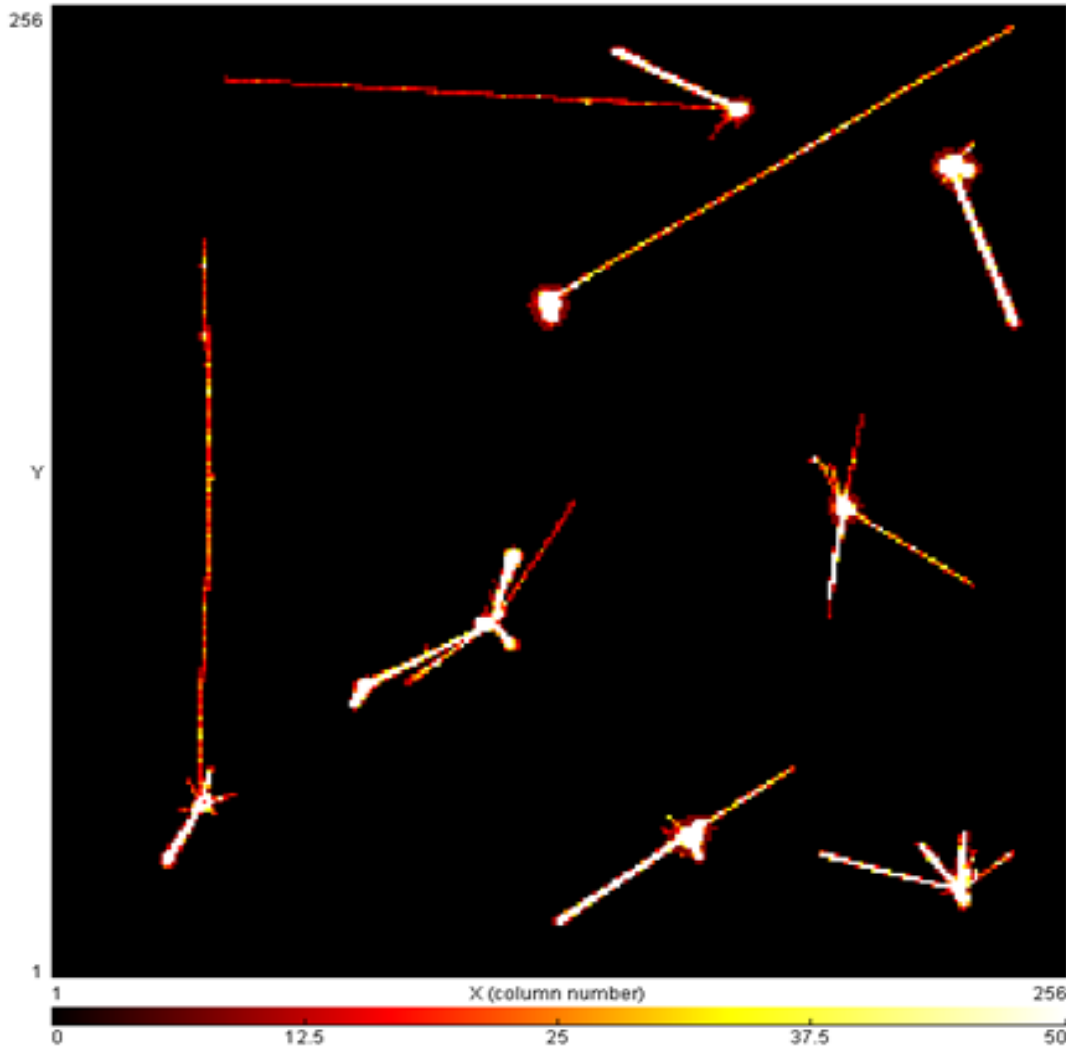




Timepix/ESA Proba-V: LEO space radiation @ 820 km



LETPs: Energetic light charged particles (II) + nuclear interactions

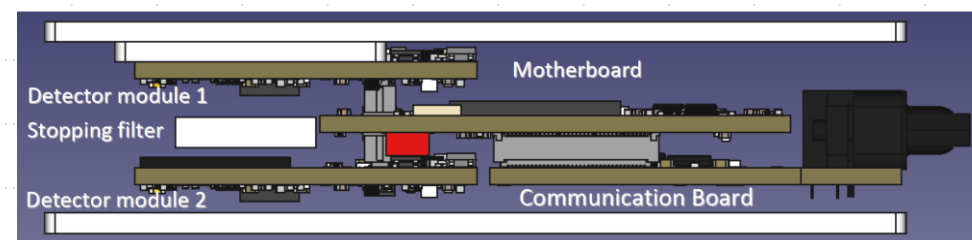
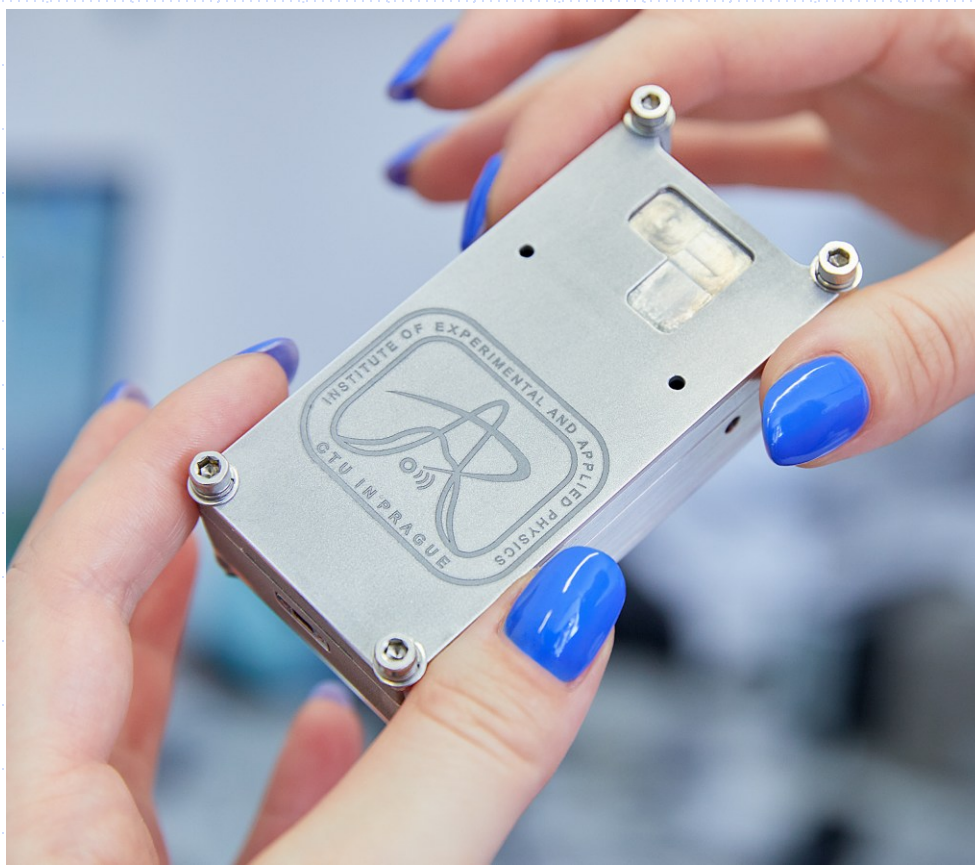




HardPix Radiation Spectrometer based on Timepix3 pixel detector



- ◆ Modular design with 1 or 2 Timepix3 layers, COTS or RadHard processor, RS422, CAN or Ethernet interface
- ◆ On-board processing (Xilinx ZYNQ Dual Core A9) – energy spectroscopy, particle identification, dosimetry



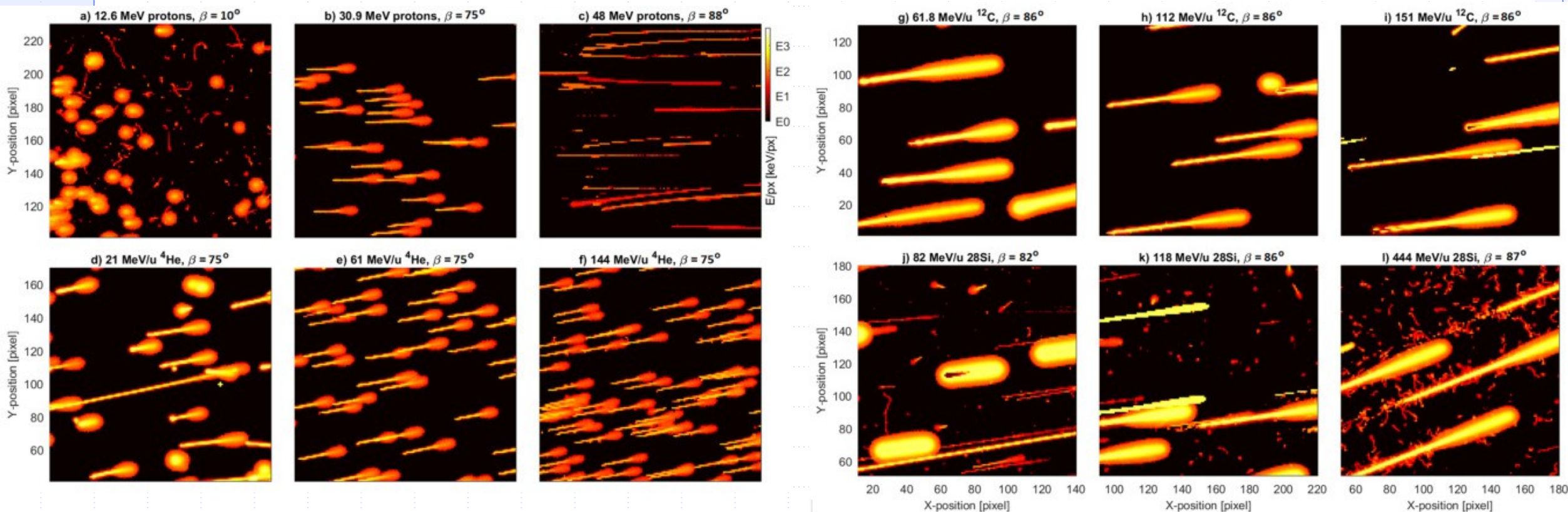
Parameters	Value
Particle types	Electrons, Protons, Ions, X-rays, Gamma-rays
Dimensions	81x40x22mm
Mass	150 g
Power consumption	1.5 – 3 W
Electrical interface	Micro Sub-D (RS-422)
Operational temperature	-20°C to 70°C
Readout chip detector	Timepix3
Sensor area	14 x 14 mm
Sensor material and thickness	Silicon 500 μ m
Voltage	5V or 28V
Energy thresholds	3 keV



HardPix Radiation Spectrometer – protons and ions



Proton and ion energy spectroscopy using energy deposition curves



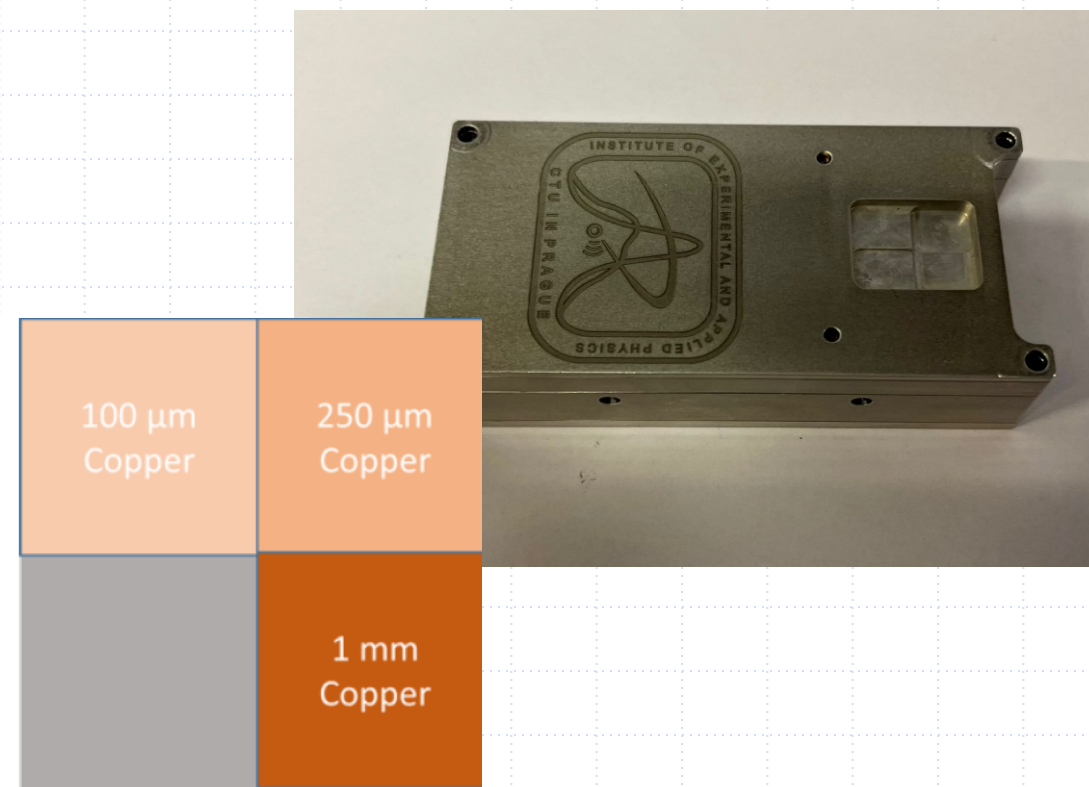
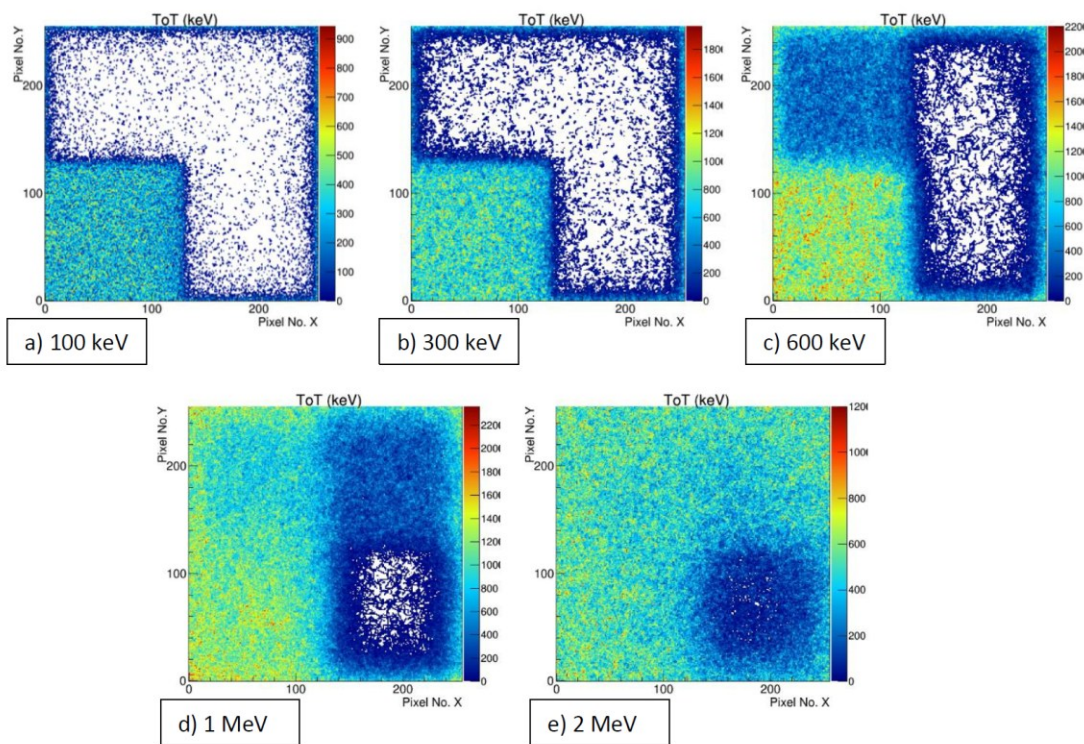
Credit: C. Granja et al., (2018), Resolving power of pixel detector Timepix for wide-range electron, proton and ion detection, <https://doi.org/10.1016/j.nima.2018.08.014>



HardPix radiation spectrometer – Electron spectroscopy



- ◆ Previous Timepix missions measured only deposited energy
- ◆ HardPix is equipped with stopping filters and provide kinetic energy of particles in several energy bins





HardPix modularity – ease of accommodation



Very minor delta development needed between missions.

Name	Interface	Note
Hardpix-SWIMMR1/2	RS422, +5V	1xTimepix3
Hardpix-ERSA	RS422, +25V	1xTimepix3
Hardpix-CASSINI	UART, +5V	2xTimepix3
Hardpix-Ethernet	Ethernet PoE	2xTimepix3

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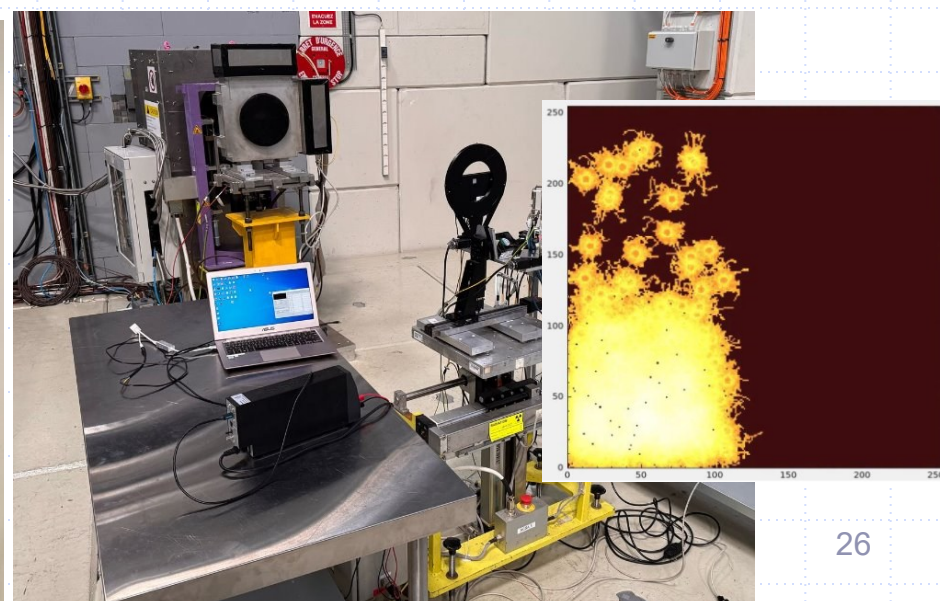
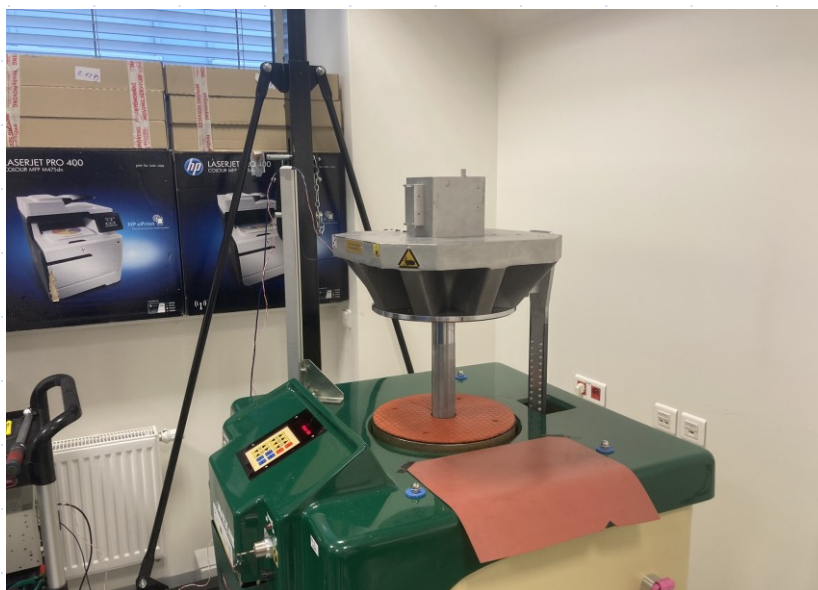
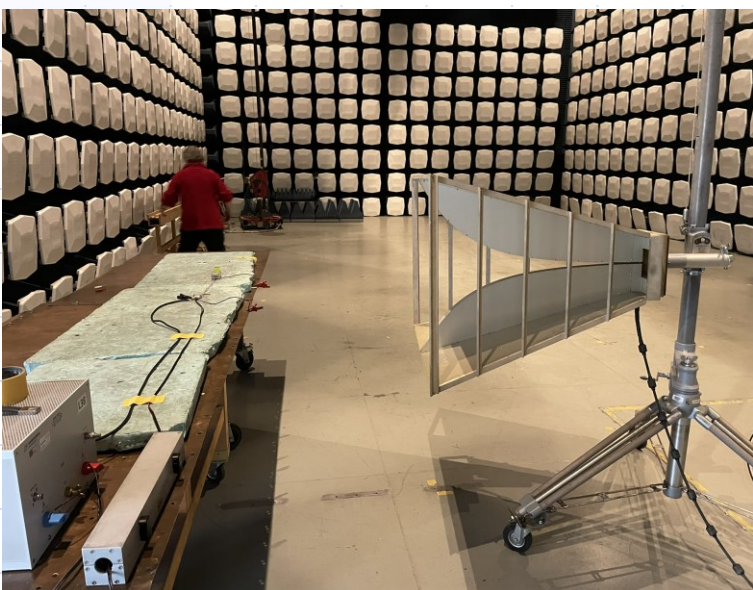
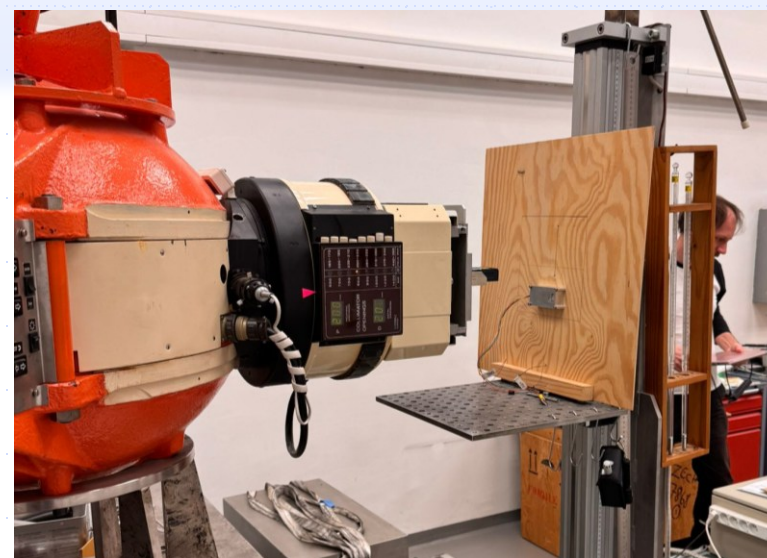
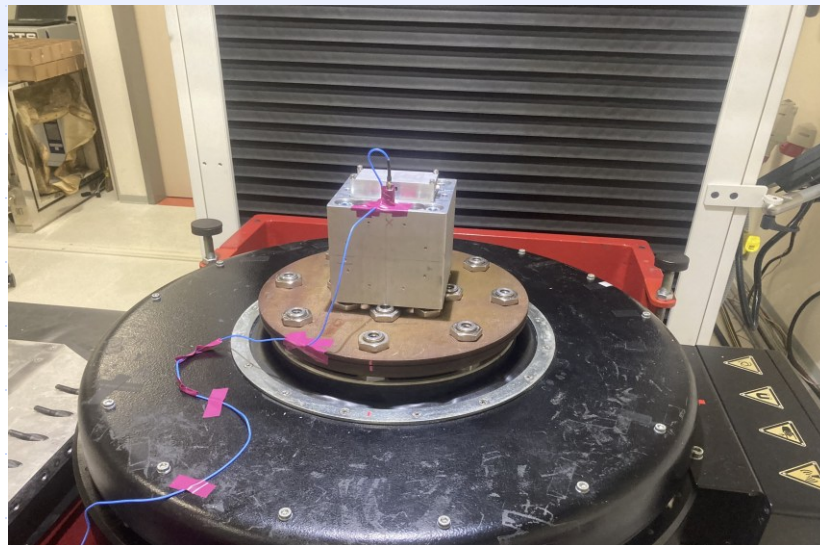
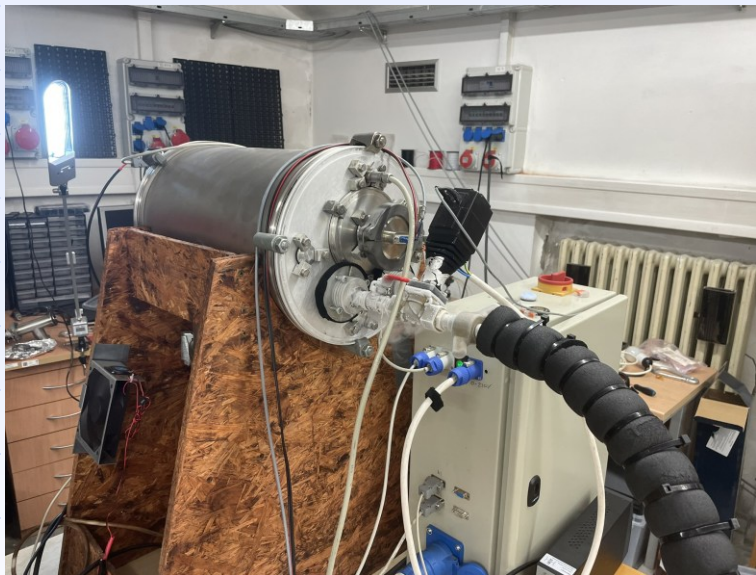




Final HardPix passed qualification campaign



TVAC (-40 to 70 C), vibrations, shocks, EMC, irradiation (30 krad Co60 + high intensity Lead (340GeV))

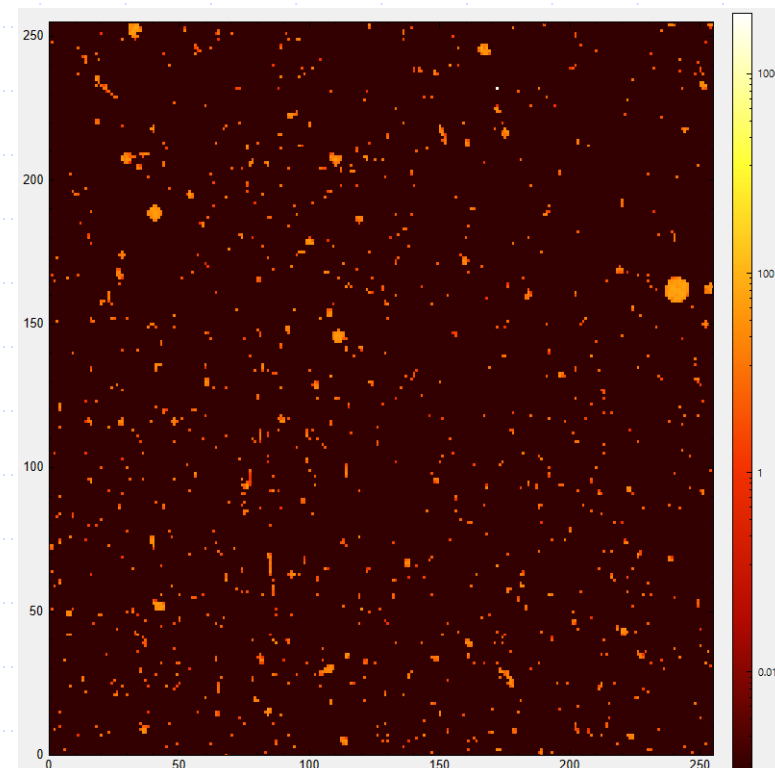
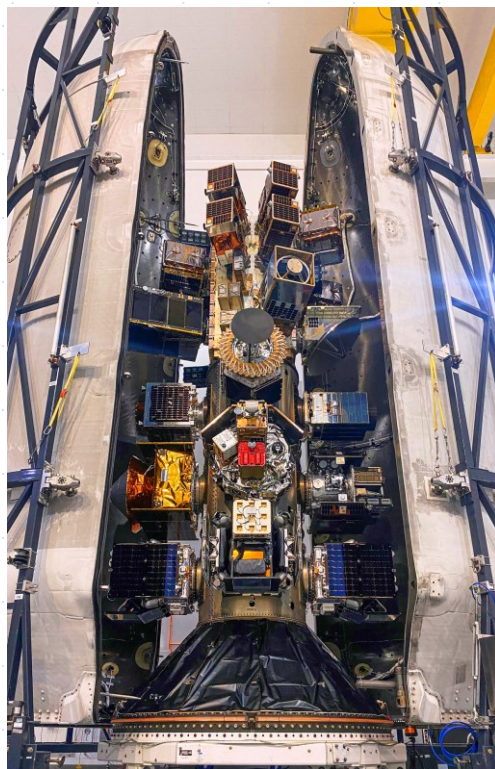




HardPix space heritage - SWIMMR1 *launched in 2023*



- HardPix prototype (single layer, no onboard processing) launched to space in June 2023 onboard D-Orbit ION satellite as part of UKRI STFC SWIMMR (Space Weather Instrumentation, Measurement, Modelling and Risk) programme

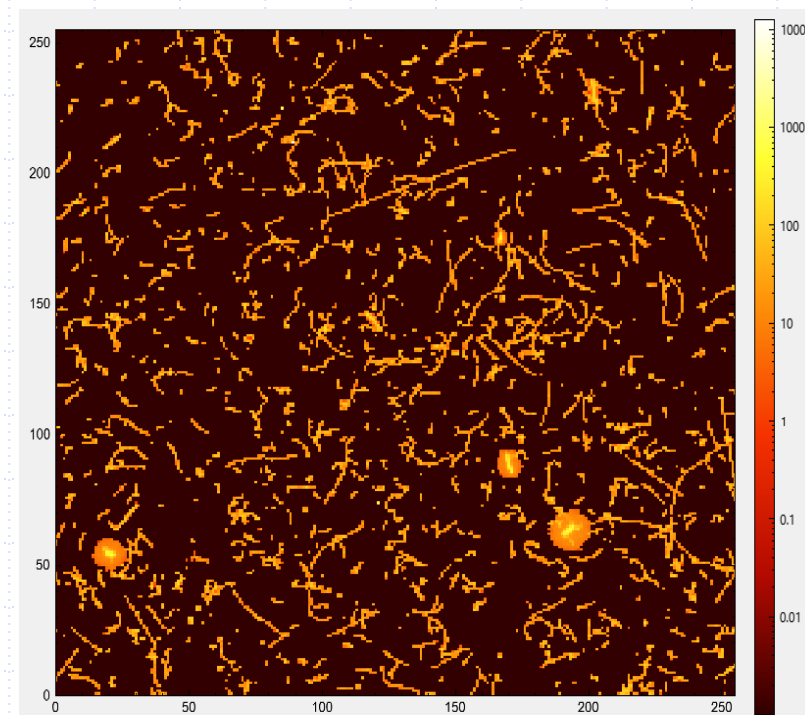
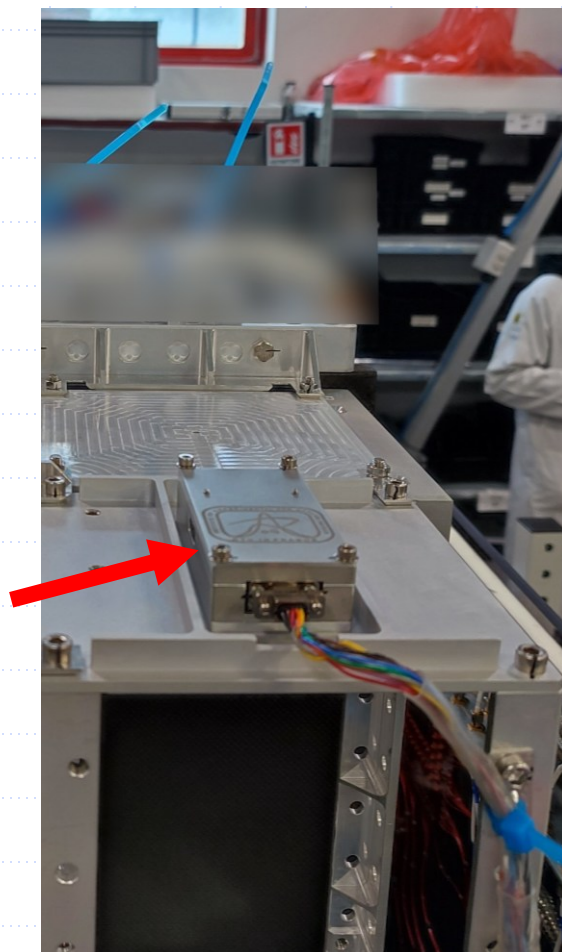
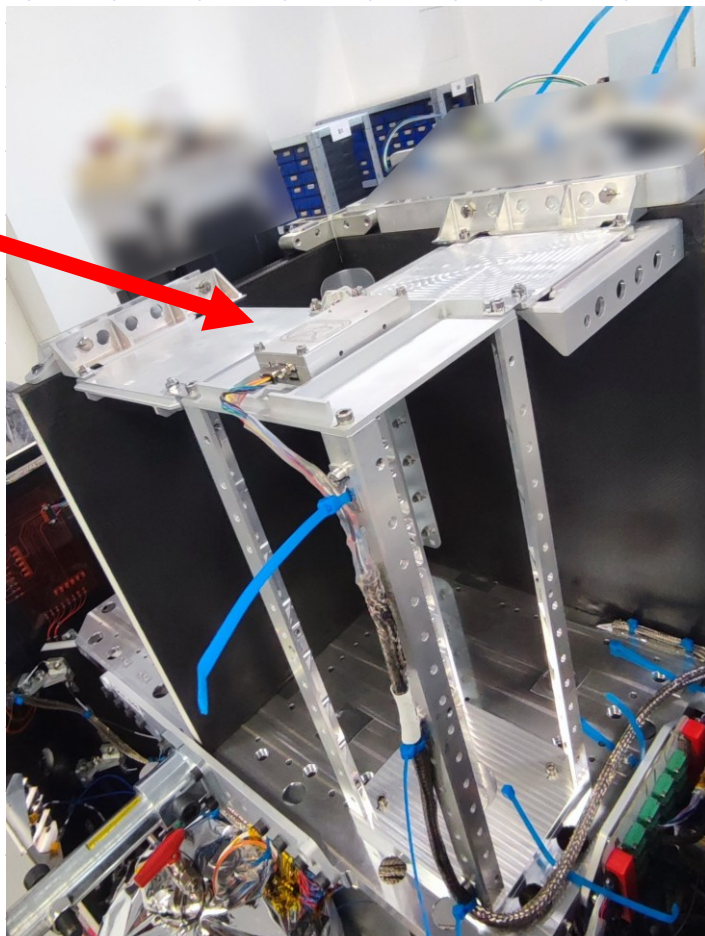


Credit: SpaceX, D-Orbit



HardPix space heritage - SWIMMR2

launched 1/2025 – apogee 1200 km

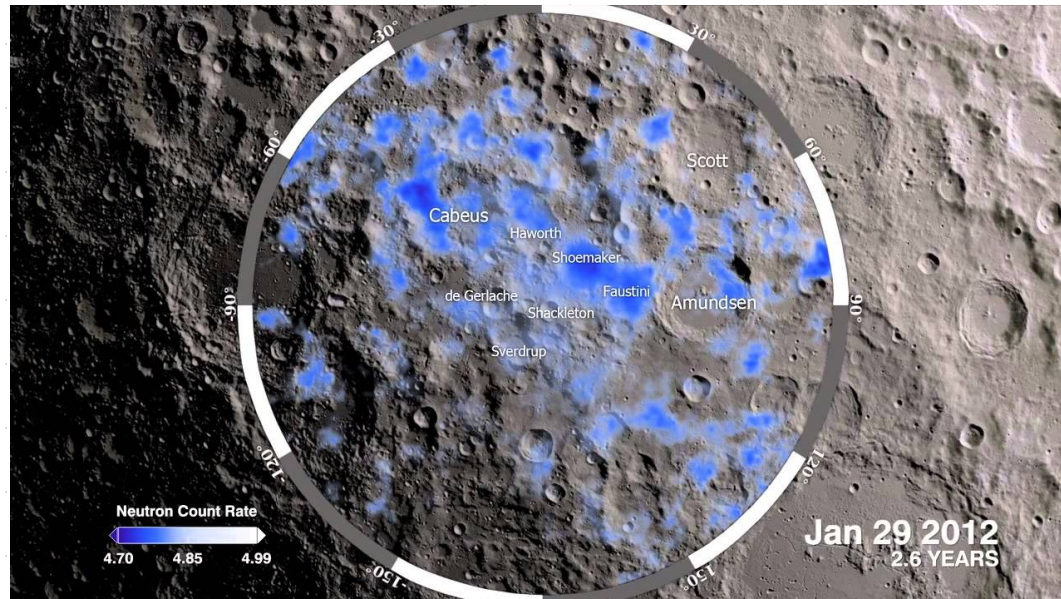




Why do we look for water on the Moon?

Because bringing it from Earth is very expensive
(~\$1M/1kg of cargo on lunar surface)

- ◆ making long-term lunar habitation feasible
- ◆ drinking water, growing plants
- ◆ breathable oxygen
- ◆ rocket fuel

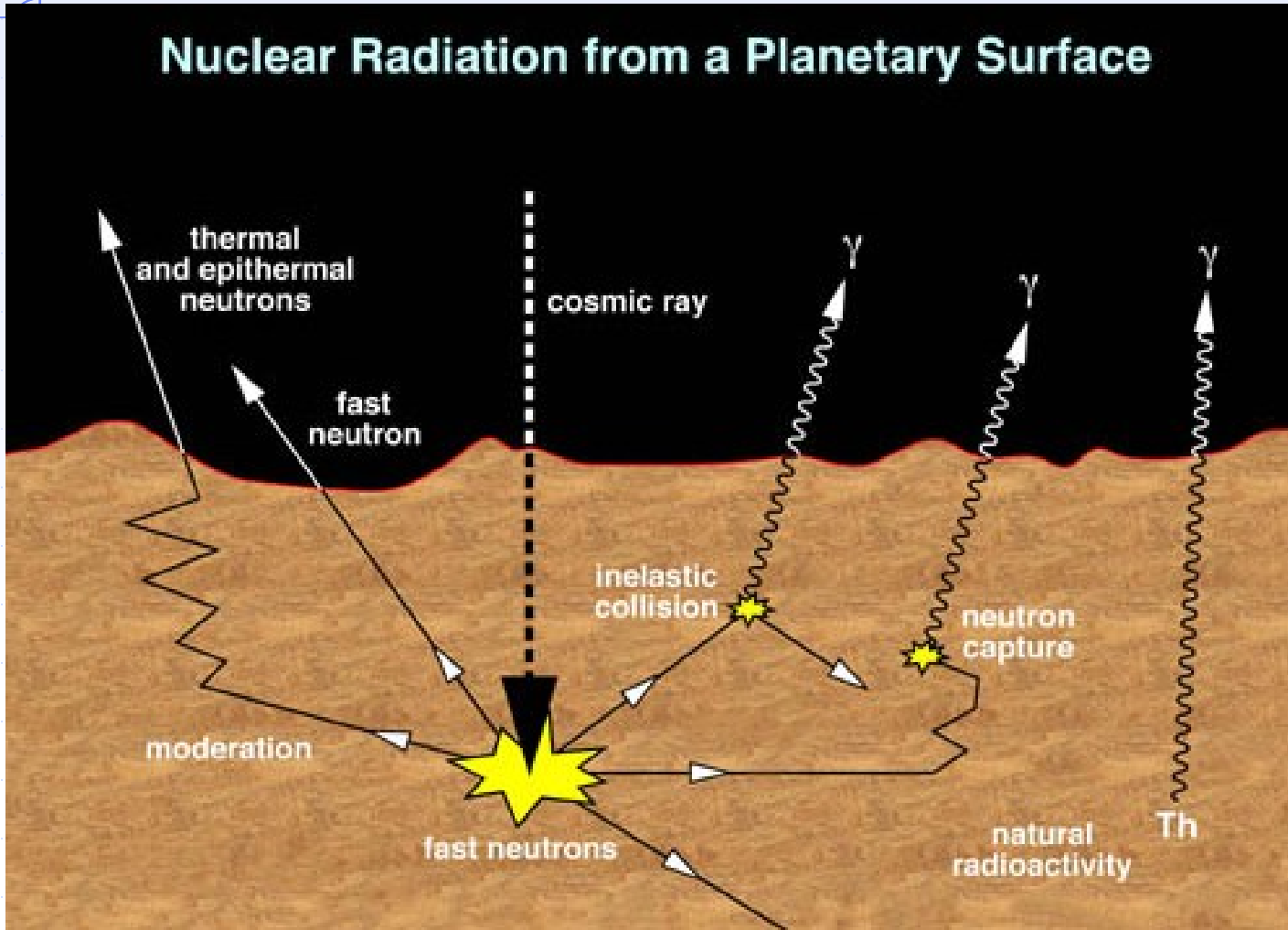


- There is no water on lunar surface as it would sublime.
- But orbital probes show possible water deposits in the form of ice in regolith subsurface (<1 m depth) in permanently shadowed regions inside craters at lunar poles.
- However, these maps have spatial resolution of 100 km, for extraction we need precise localization using detectors directly on lunar surface.
- Drilling blindly is inefficient – neutron spectrometers are used to map water deposits.



How does it work?

Neutrons on lunar surface



- Galactic cosmic rays (GCRs) constantly interact with the nuclei in the lunar subsurface and produce high energy neutrons.
- Neutrons are always created fast, as they have to be liberated from the atomic nuclei, which are generally strongly coupled systems of protons and neutrons.
- These neutrons scatter in the regolith and lose energy or are absorbed by neutron capture reactions
- Hydrogen in water molecules moderates the neutrons and therefore changes the neutron spectrum
- *Timepix detectors can be adapted to for detection of fast neutrons and moderated neutrons*



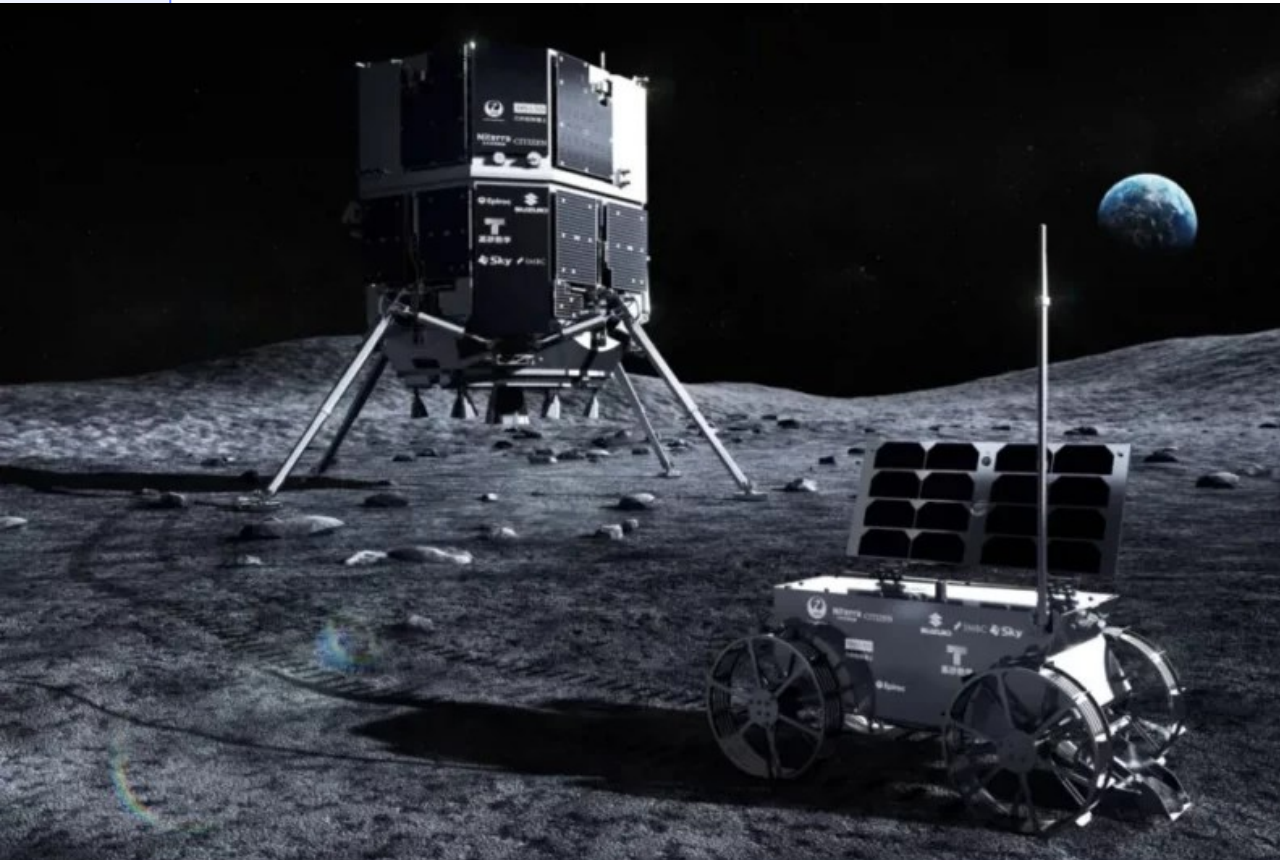
MAGPIE (Neutron HardPix payload)



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The Mission for Advanced Geophysics and Polar Ice Exploration (MAGPIE) led by ispace Europe (LU).
ispace rover with **Neutron HardPix**, Ground penetrating radar, drill+mass spectrometer planned to be launched in 2028.

First ispace lander failed to land in April 2023.
Second attempt on June 5th 2025 failed as well.



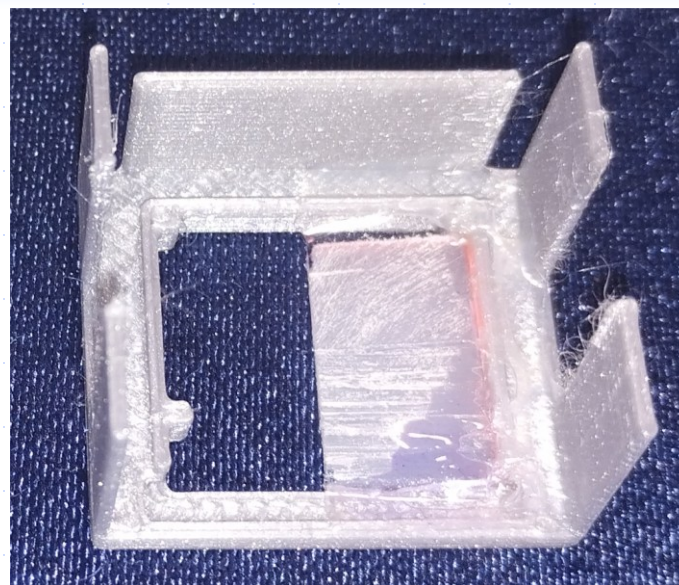
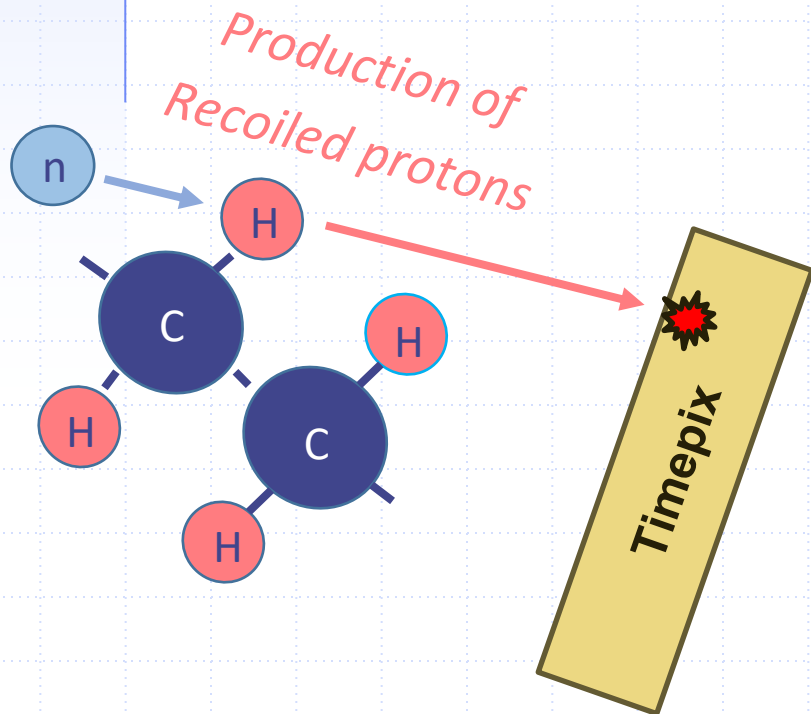
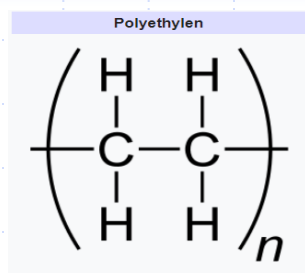
ispace



Detection of Fast Neutrons

Adaptation of Timepix pixel detector

Application of PolyEthylene Converter



- PE (PolyEthylen) *represents hydrogen rich material*
- *Recoiled protons are clearly recognized in the pixel detector as heavy blob tracks*



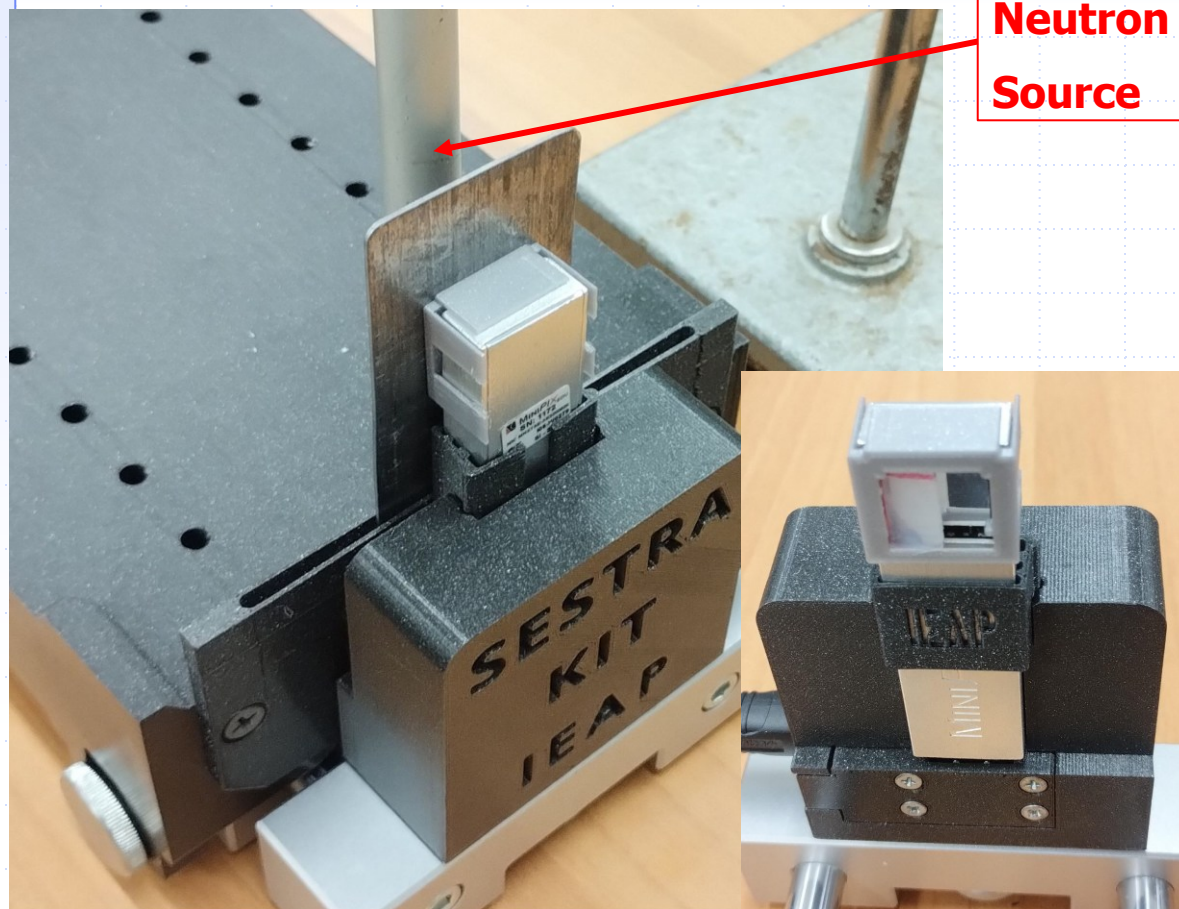


Detection of Fast Neutrons using pixel detector

Bare vs Adapted sensor covered with PE Converter



Neutron
Source

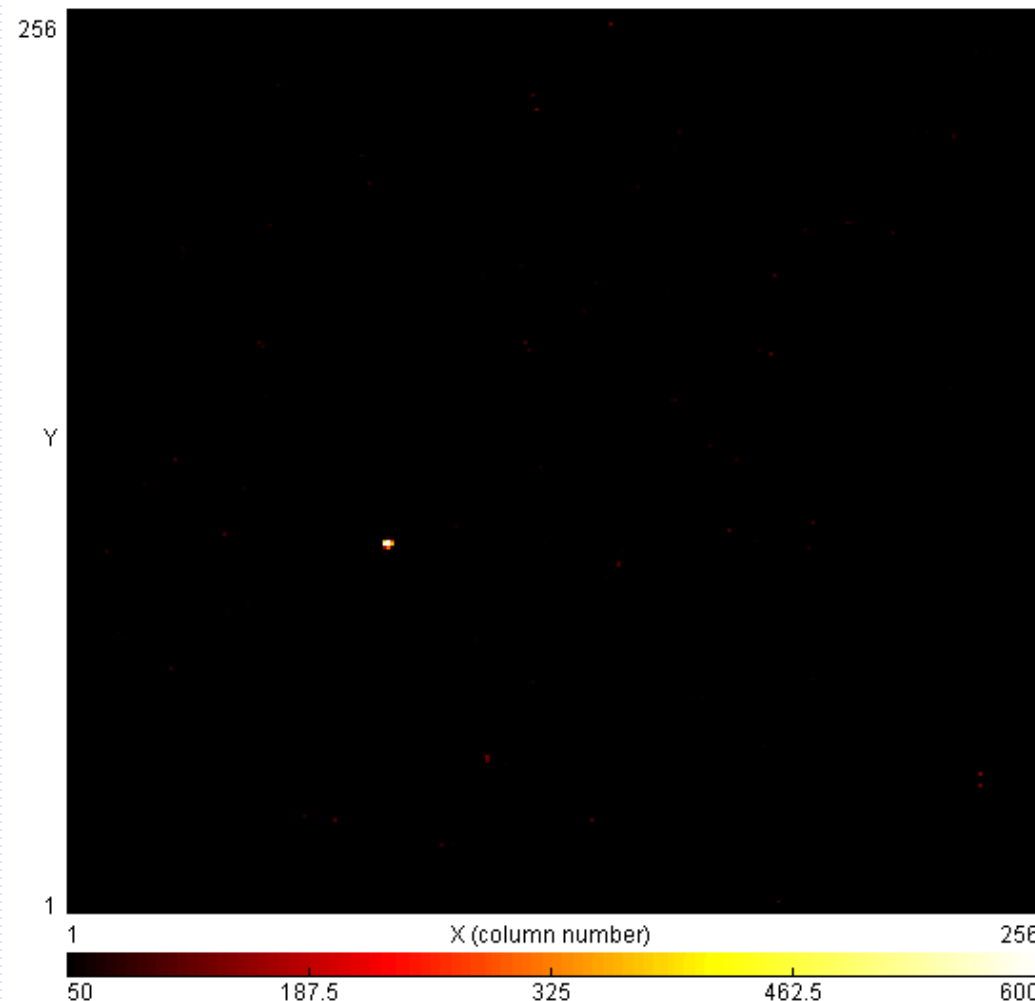


*30 mCi AmBe neutron source used
Replay of 4 min long measurement*

Adapted Sensor



Bare Sensor





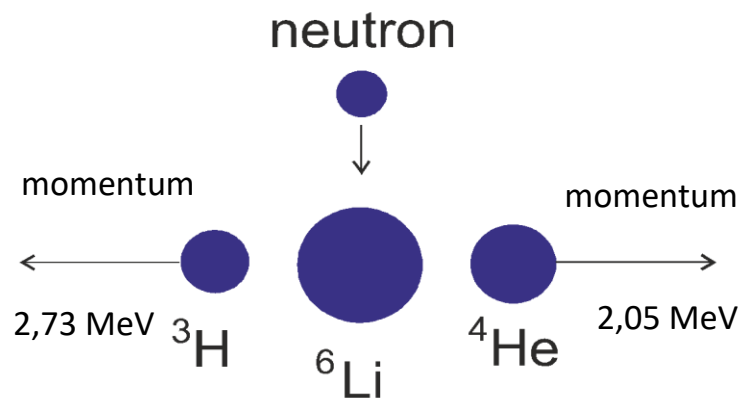
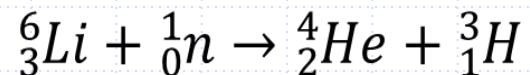
Detection of Thermal Neutrons

Adaptation of Timepix detector

Application of LiF Converter



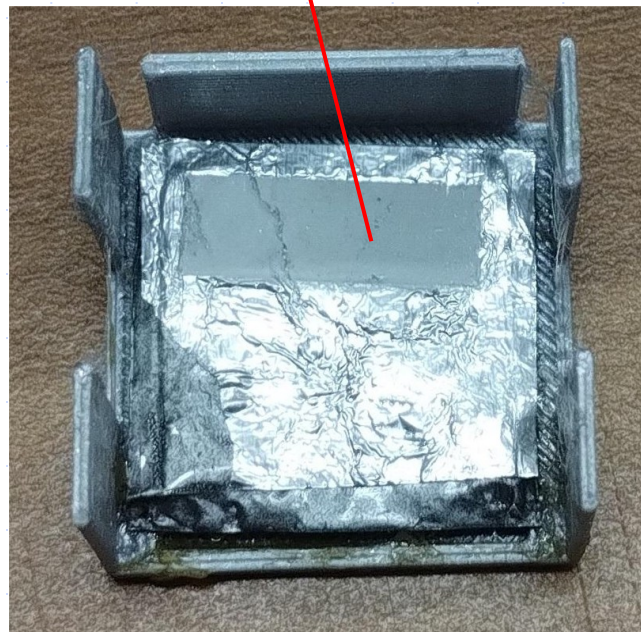
Nuclear reaction of neutrons with lithium



Heavy charged ions are produced in the nuclear reaction

LiF Converter

3,6 mg/cm²



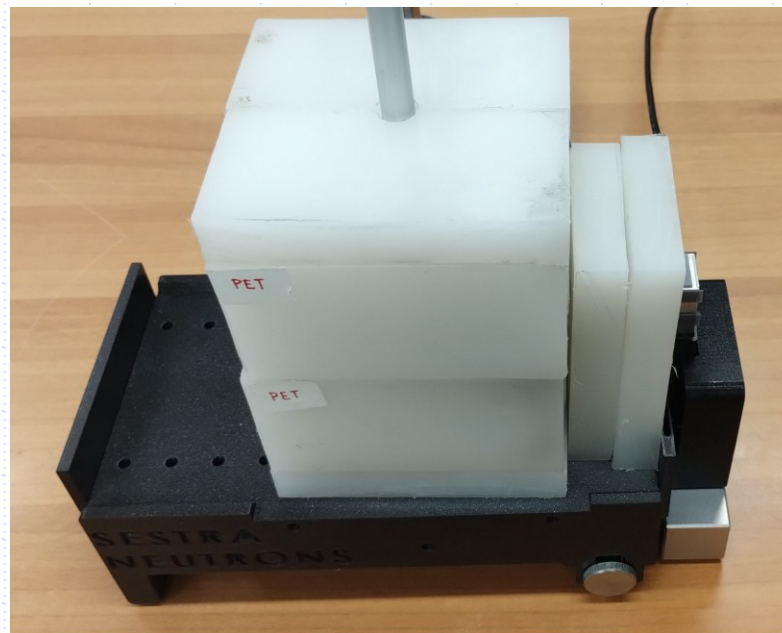
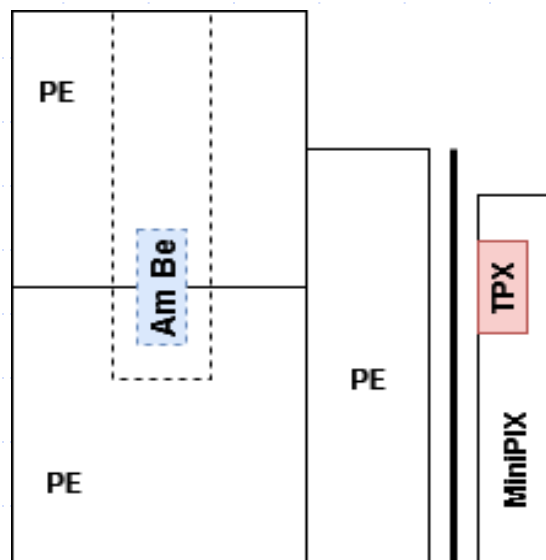
Tritons and alphas are clearly recognized in the pixel detector as heavy blob tracks



Thermal Neutron Detection Arrangement of Experiment Set-up



- + PE moderating blocks
- + Neutron source (AmBe) inserted inside moderator
- + Timepix detector placed side-by
- + Pb plate for gamma shielding (Am)



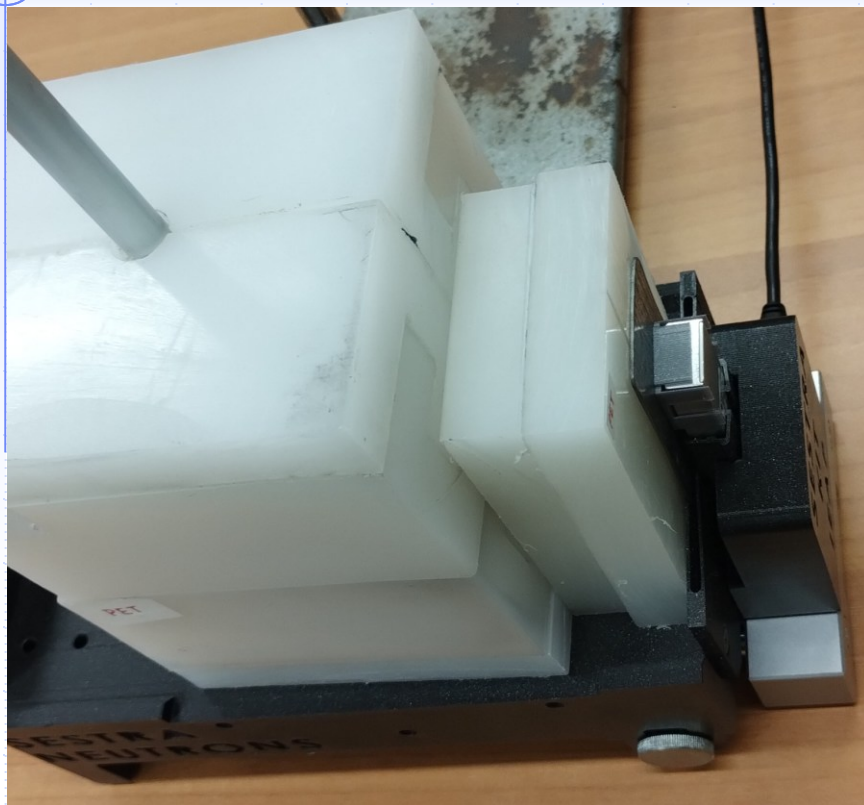
Note> PE material is rich in hydrogen (paraffin, water would also work well in neutron moderation)



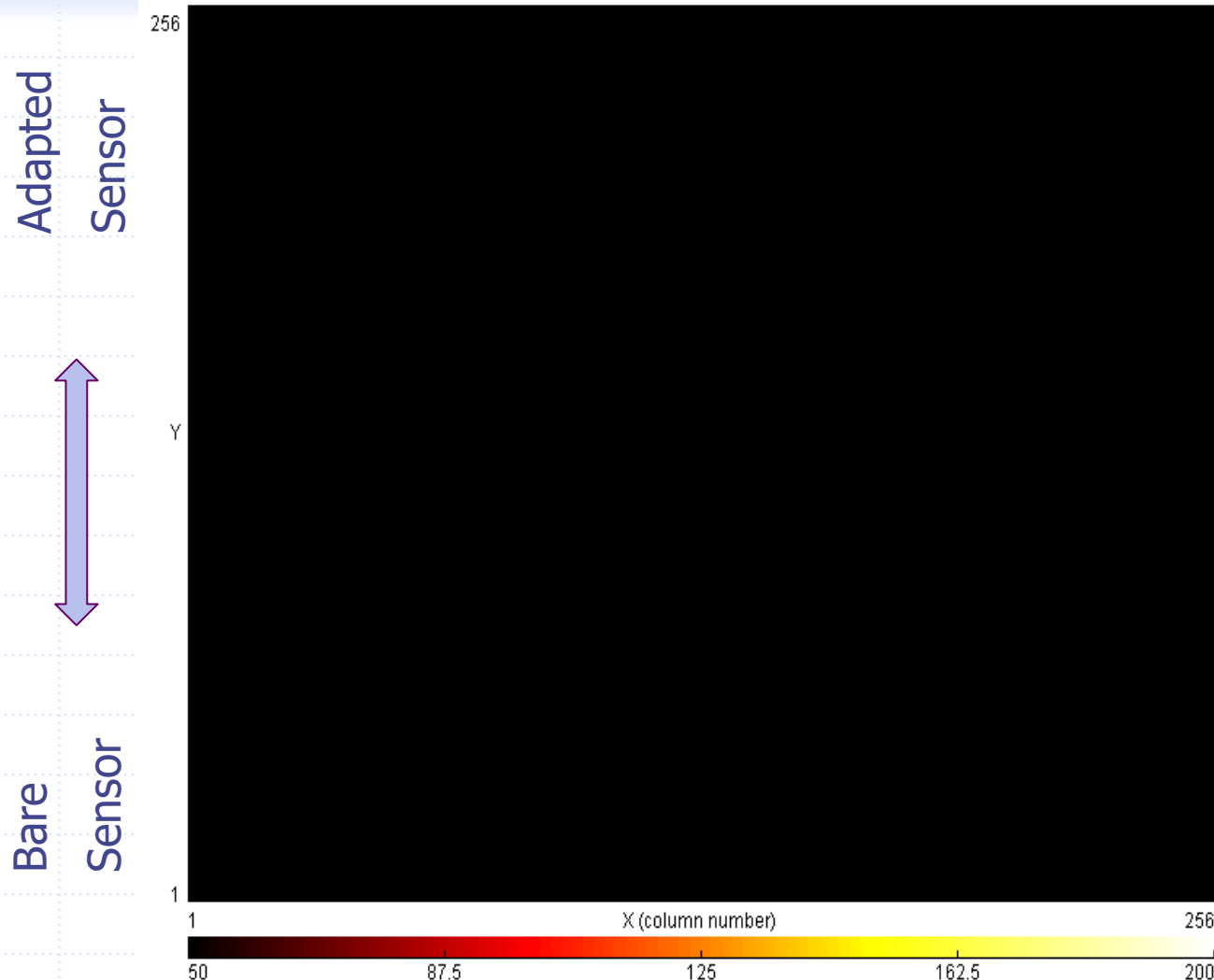
Detection of Thermal Neutrons using pixel detector



...Bare vs Adapted sensor covered with LiF Converter

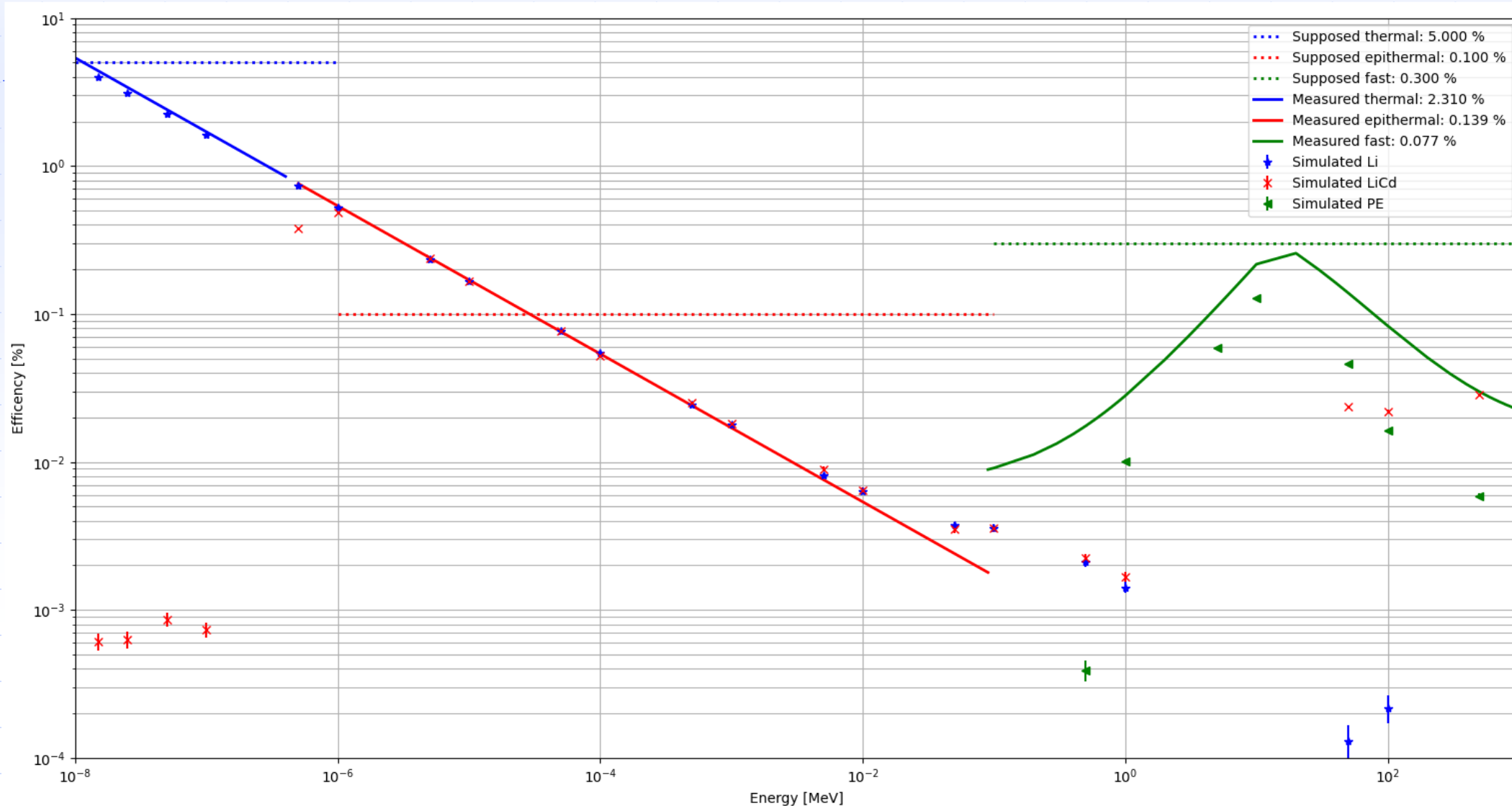


*30 mCi AmBe neutron source used
Replay of 4 min long measurement*





Neutron HardPix - detection efficiencies





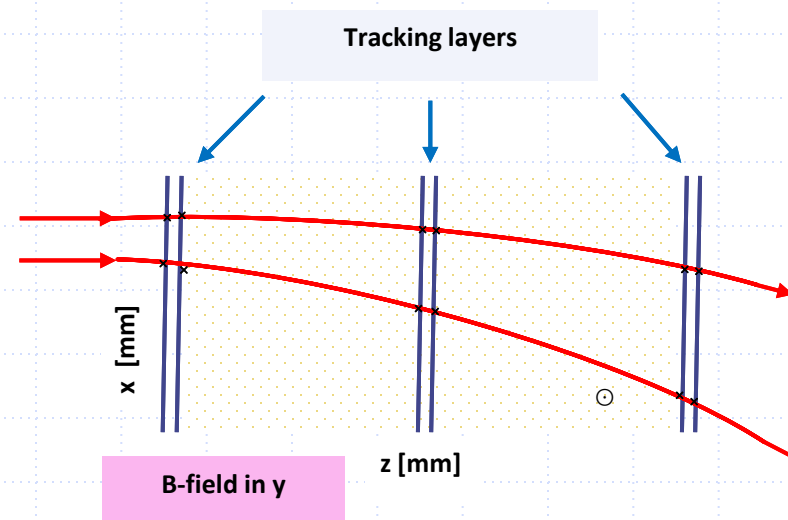
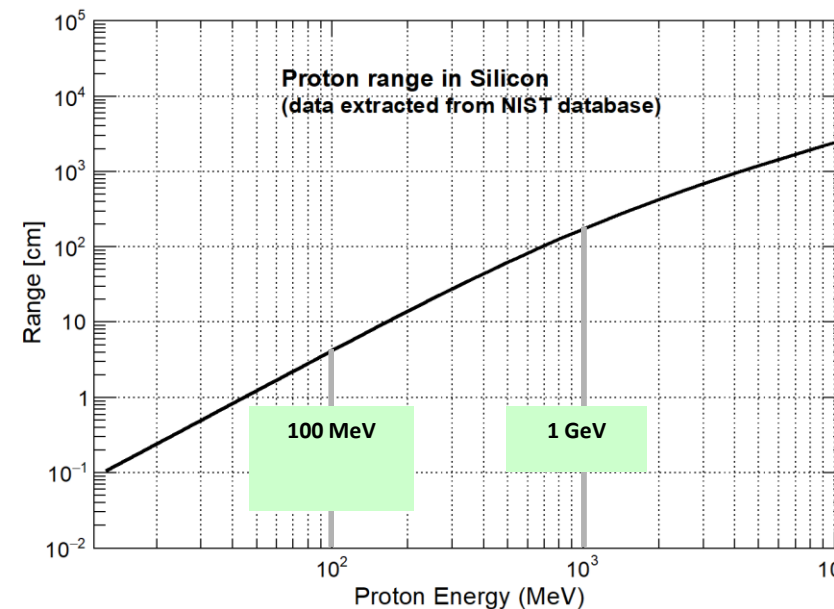
Challenge of measuring GeV particles



- **Energy of GeV particles cannot be measured by the ΔE -E method**
 - E.g. only ~4 cm of Silicon needed to stop 100 MeV protons with ionization energy loss
 - **To stop 1 GeV protons 170 cm of Silicon is needed!**
 - Silicon nuclear interaction length = 46.52 cm
 - More likely to produce a **shower of secondary particles** (including the dangerous neutrons)
 - Calorimeter method: **very heavy, with bad resolution**
 - Similarly relativistic electrons shower in thick material
- **PAN: use a magnetic spectrometer to measure the bending of charged particles in the B-field $\Rightarrow$ rigidity (p/Z)**
 - then infer the momentum and energy with independently measured particle charge Z

Magnetic spectrometer = magnetic field + tracking detectors

Since particles of different masses and charges will have different rigidities for the same momentum, this allows for the separation and identification of different particle species.



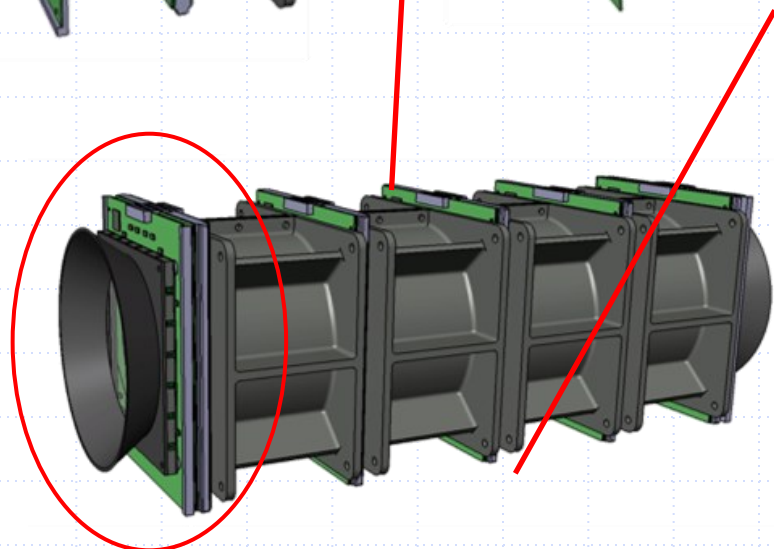
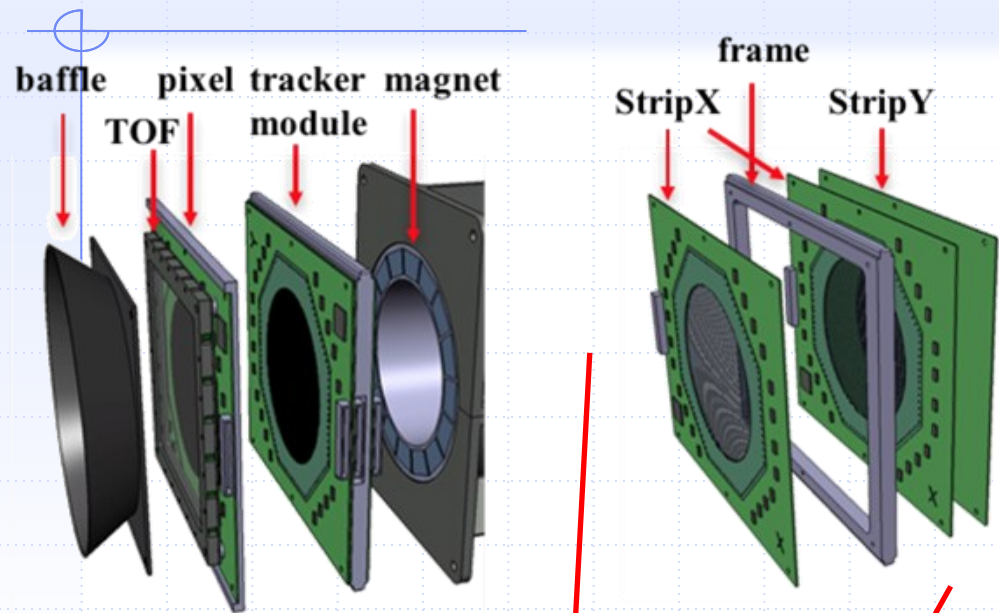


PAN instrument: Technical viewpoint

<http://www.pan-space.eu/>

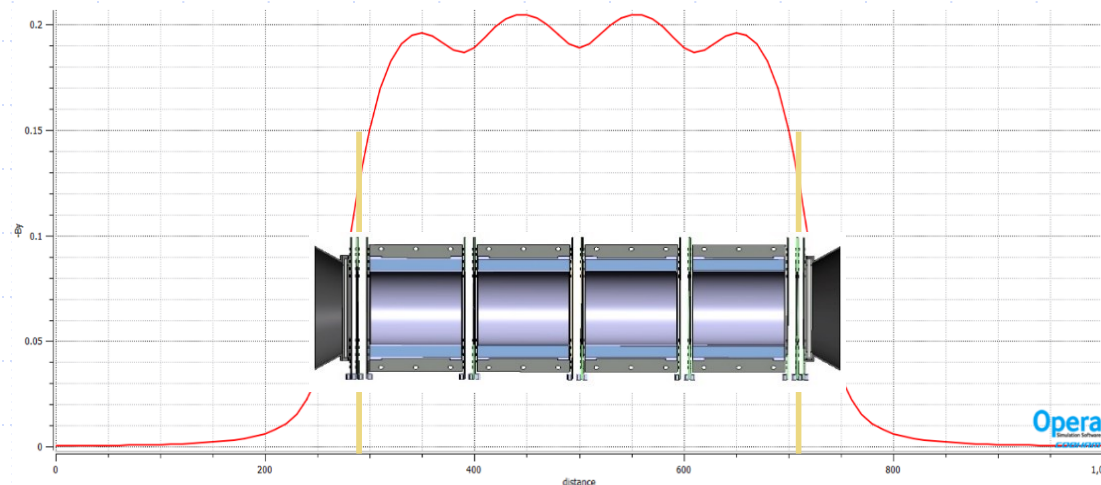


Penetrating Particle Analyzer



- ◆ Light
- ◆ Low power
- ◆ Symmetric: Measured particles coming in from both ends
- ◆ 4 Halbach permanent magnet sectors, each ($\phi = 10$ cm, $L = 10$ cm)
 - Dipole magnetic field of ~ 0.2 Tesla

Magnetic field profile:





Scientific objectives of mini.PAN



Penetrating Particle Analyzer



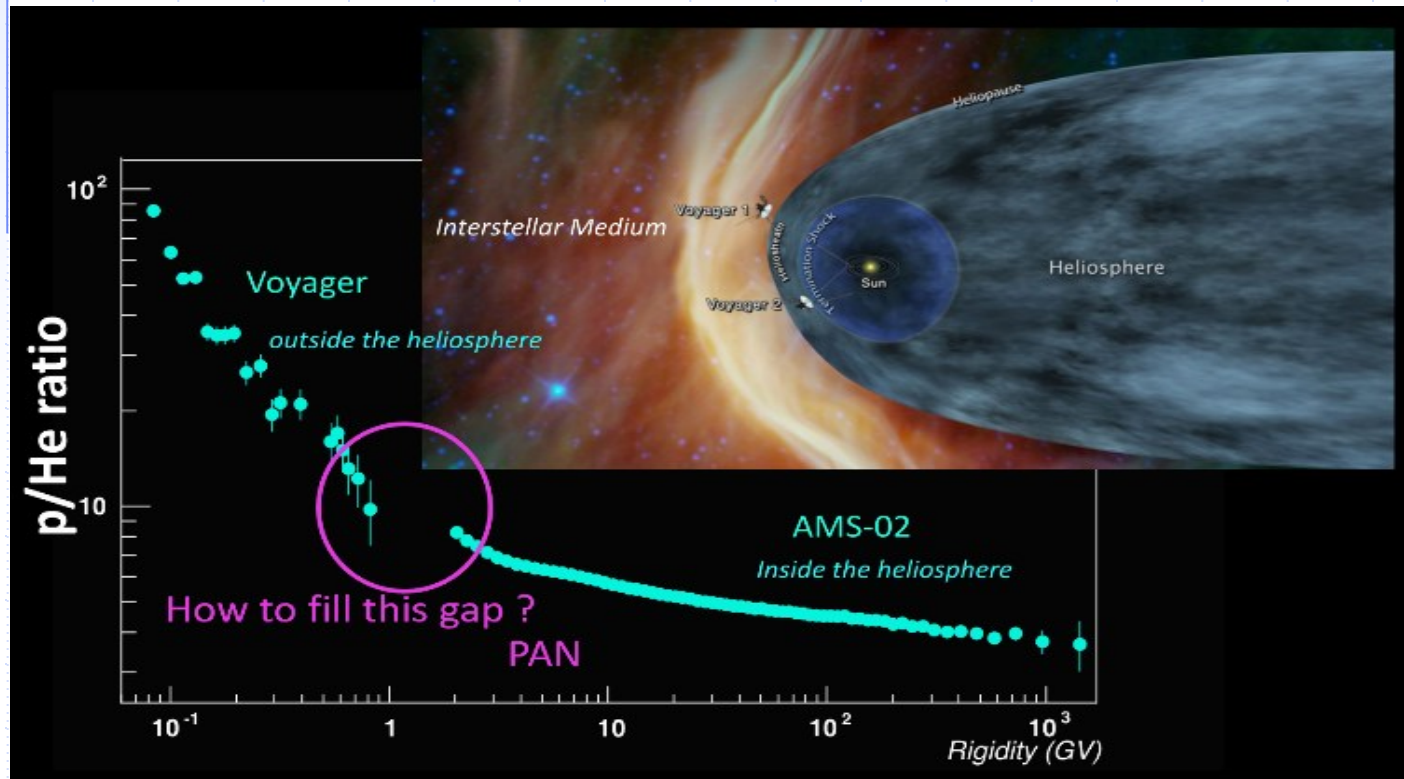
Precisely measure and monitor flux, composition, and direction of penetrating particles (up to ~ 20 GeV/nucleon)

Galactic cosmic ray (GCR) physics:

- Understand origin, acceleration mechanisms and propagation properties
- Antimatter content

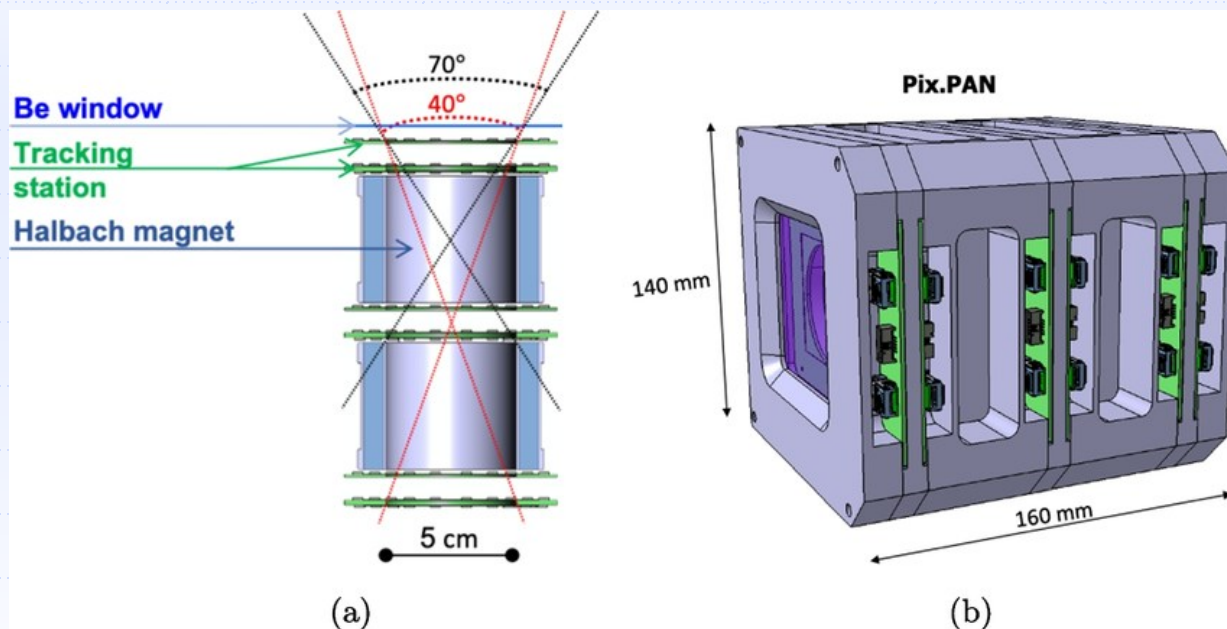
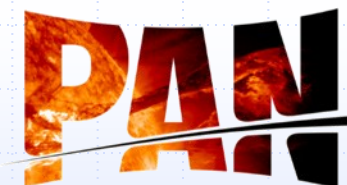
Dosimetric aspect:

- GCR major source of dose for astronauts on (interplanetary) missions

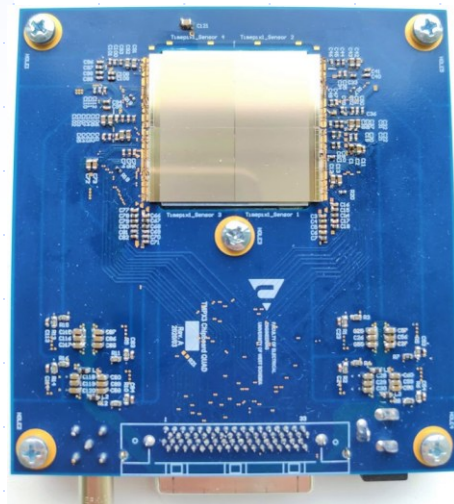




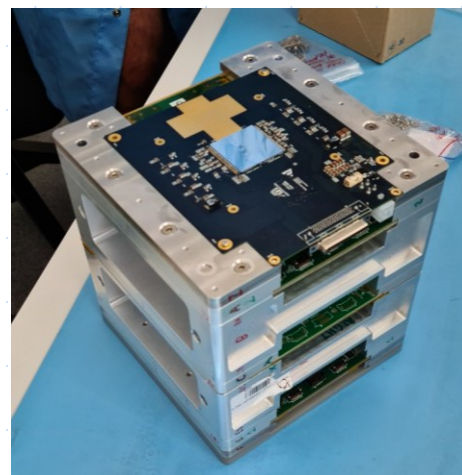
Pix.PAN



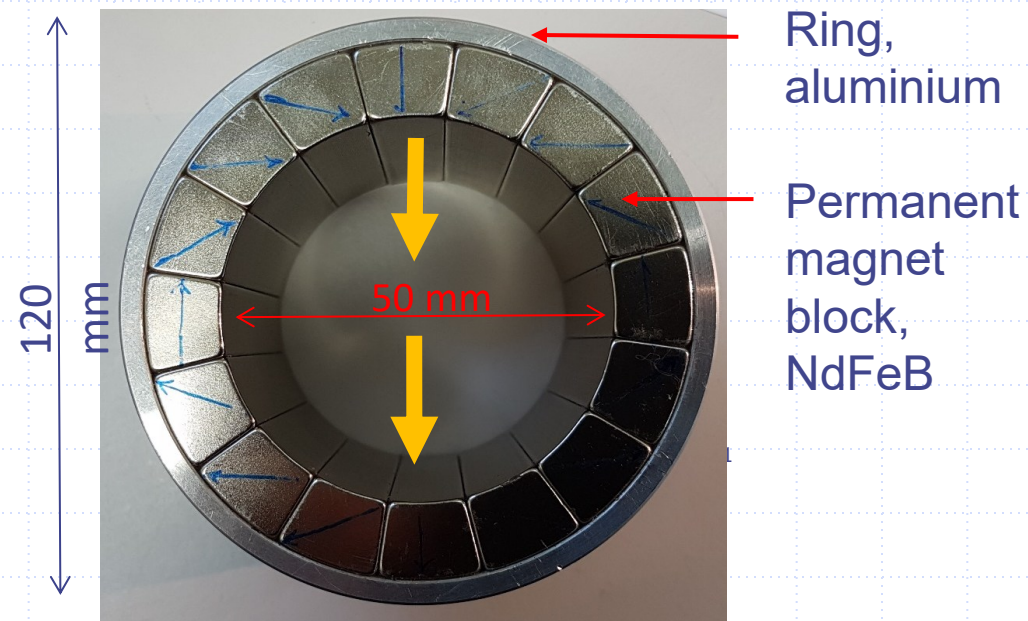
- ◆ Pix.PAN using Timepix4 as tracking layers
- ◆ Planned to fly onboard LUNPAN mission outside of Earth's magnetosphere



Pixel module



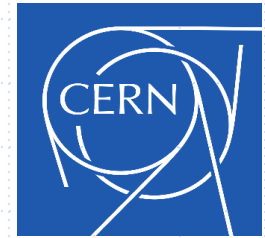
Pixel module integration



Magnet sectors



Thank you for your attention



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